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PASSI CITY COLLEGE
City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

ECOPOINTS: A SMART WASTE COLLECTION AND REWARD REDEMPTION SYSTEM

A Capstone Project Presented to the Faculty of the School of Information and Communication Technology Passi City College
City of Passi, Iloilo

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CHAPTER I

INTRODUCTION

Project Context

Background of the Study

Waste management is a pressing global issue, with improper disposal contributing to environmental pollution, public health concerns, and inefficient recycling processes. Traditional waste collection methods often lack incentives for individuals to actively participate in proper waste segregation and disposal. As urbanization and consumerism increase, there is a growing need for smart and sustainable waste management solutions that encourage responsible behavior while promoting environmental conservation.

EcoPoints: A Smart Waste Collection and Reward Redemption System is designed to address this issue by integrating technology and incentives to promote proper waste disposal. This system utilizes sensors and microcontrollers, such as Arduino and ESP32, to collect real-time data on waste disposal activities. The collected data is then transmitted to an online platform, where users can monitor their waste disposal habits and earn reward points, known as EcoPoints. These points can be redeemed for various incentives, encouraging sustainable practices among communities.

The proposed system aims to bridge the gap between waste disposal and community engagement by leveraging digital solutions. With the implementation of an interactive web dashboard, users can track their contributions, access waste collection data, and participate in reward programs through an easy-to-use interface. By promoting responsible waste management through gamification and incentives, EcoPoints seeks to foster a culture of sustainability and environmental awareness.

This study explores the development and implementation of the EcoPoints system, assessing its potential impact on waste collection efficiency and public participation in sustainable practices. It seeks to determine how integrating smart technology with a reward-based model can improve waste management behaviors, reduce landfill waste, and contribute to a cleaner environment.

Statement of the Problem

Waste mismanagement remains a significant environmental and societal issue, leading to pollution, inefficient recycling, and the excessive accumulation of waste in landfills. Despite existing waste collection initiatives, many individuals lack motivation to properly segregate and dispose of waste, resulting in poor participation and

engagement. Traditional waste management systems often do not provide incentives for users, making it difficult to encourage consistent and responsible waste disposal behaviors.

The EcoPoints: A Smart Waste Collection and Reward Redemption System aims to address these challenges by integrating technology with an incentive-based approach. However, several key problems need to be investigated:

- Lack of Public Engagement in Proper Waste Disposal – How can an incentive-based system encourage individuals to actively participate in proper waste segregation and disposal?

- Inefficiency in Monitoring Waste Disposal Activities – How can real-time data collection through sensors and microcontrollers improve the tracking of waste disposal habits?

- Limited Awareness and Motivation for Recycling – Can a reward-based system effectively promote sustainability and increase awareness of proper waste management practices?
- Challenges in Integrating Technology into Waste Management – What are the technical and operational

limitations of implementing a smart waste collection and reward system?

This study seeks to explore these issues by developing and evaluating the EcoPoints system, determining its effectiveness in improving waste management practices, and analyzing its potential impact on environmental sustainability.

Project Description

The EcoPoints system is an innovative solution designed to modernize waste management by integrating smart technology with a reward-based recycling initiative. It focuses on crucial aspects of waste tracking, user participation, and reward redemption, making waste disposal more engaging and efficient. By automating these processes, the system ensures greater accuracy, transparency, and efficiency in managing waste collection and incentive distribution. This initiative marks a significant step toward environmental sustainability while utilizing digital solutions to encourage responsible waste disposal.

The EcoPoints system provides a centralized platform for tracking and managing waste collection activities and user participation. Users can monitor their waste contributions,

earn EcoPoints based on the type and quantity of waste they dispose of, and redeem their points for rewards. This approach gamifies the waste disposal process, increasing public engagement and promoting a culture of recycling.

Although the system primarily focuses on waste collection, user engagement, and reward management, it does not cover waste logistics (such as pickup scheduling or route optimization), keeping its scope clear and manageable. Despite some limitations, such as predefined point allocation for waste types and structured reward redemption methods, the EcoPoints system represents a significant advancement in waste management innovation.

The EcoPoints system highlights the potential of technology-driven sustainability efforts. By automating waste tracking and incentivizing responsible behavior, it empowers individuals to contribute to a cleaner environment. Furthermore, the project serves as a foundation for future enhancements and as a valuable resource for researchers and organizations exploring smart waste management solutions.

Objectives of the Study

The EcoPoints: A Smart Waste Collection and Reward Redemption System aims to establish and implement an efficient and engaging waste management system that encourages responsible waste disposal through a reward-based approach.

General Objective

The primary aim of this study is to develop and enhance the functionality of the EcoPoints System to promote environmentally sustainable waste disposal. This system focuses on accurately recording waste collection data, assigning reward points based on waste type and quantity, and ensuring seamless redemption of rewards. By evaluating the current waste disposal practices, the study aims to identify areas for improvement, such as minimizing manual processes, enhancing user engagement, and ensuring efficient tracking of waste contributions. The goal is to create a system that incentivizes proper waste management while providing users with a reliable platform for monitoring their contributions and rewards.

Specific Objectives

1. Provide an accessible platform for users to track their waste contributions and earned EcoPoints.
2. Enhance efficiency in waste tracking by automating the collection and processing of waste disposal data.

3. Ensure accuracy and transparency in reward allocation based on waste type.
4. Encourage user participation by integrating a reward-based system that motivates responsible waste disposal.
5. Promote environmental awareness by educating users on the benefits of recycling and proper waste segregation.
6. Ensure data security and reliability by implementing measures to protect user information and system integrity.

Scope and Delimitation

The study focuses on the development, implementation, and evaluation of the **EcoPoints: A Smart Waste Collection and Reward Redemption System**, which is designed to improve waste disposal practices through technology and incentives. The system utilizes **ESP32 microcontrollers** to collect real-time waste disposal data and transmit it to an online platform. A web-based dashboard is developed to allow users to monitor their waste contributions, track collected data, and redeem **EcoPoints** as rewards.

However, the study has several limitations. It does not cover waste logistics such as pickup scheduling, route optimization, or large-scale waste processing. The reward

system follows a **predefined allocation method**, meaning users can only earn and redeem points through manual process. Additionally, as the study is implemented solely within **Passi City College**, the findings may not directly apply to large-scale waste management systems. The system also relies on an internet connection for real-time data transmission, which may affect its functionality in areas with limited connectivity. Despite these limitations, the study provides valuable insights into how smart technology and gamification can enhance waste management and serves as a basis for future research and development in the field of sustainable waste management.

Significance of the Study

The **EcoPoints: A Smart Waste Collection and Reward Redemption System** is an innovative approach to waste management that integrates smart technology with an incentive-based system to encourage responsible waste disposal. This study is significant to various stakeholders, particularly students and faculty, as it promotes proper waste segregation and fosters a culture of sustainability and environmental awareness within the campus. The school administration will also benefit from the system by having an

organized and efficient waste management process, allowing for real-time monitoring of waste disposal activities.

Furthermore, local government units (LGUs) and environmental organizations may use the findings of this study as a reference for developing community-wide smart waste management initiatives, contributing to policy improvements and recycling efforts. Additionally, researchers and developers in the fields of smart waste management, IoT (Internet of Things), and sustainability can use this study as a foundation for further innovations and enhancements. By addressing common challenges in waste disposal through the use of technology and incentives, this study aims to contribute to the promotion of sustainable environmental practices, reduction of landfill waste, and increased community engagement in responsible waste management.

Definition of Terms:

- **Admin Panel:** A web-based dashboard designed for system administrators to oversee system performance, manage users, configure bins, and track usage analytics.
- **Authentication & Authorization:** Mechanisms for ensuring secure access to the system. Authentication verifies user identity; authorization defines user permissions based on roles (e.g., user, admin).
- **C/C++:** Used in programming the ESP32 microcontroller for handling sensors, RFID readers, and communication protocols.
- **Data Logging:** The process of collecting and storing data points such as timestamp, user ID, waste weight/type, and bin location each time a deposit is made.
- **EcoPoints:** A virtual point-based incentive currency awarded to users for depositing waste into smart bins. Points are used as a behavior change tool to promote sustainable habits.
- **ESP32 Microcontroller:** A Wi-Fi and Bluetooth-enabled chip used to process sensor data in smart bins and communicate with the server over a network (e.g., RESTful API via HTTP).
- **HTTP/HTTPS:** Protocol used by smart bins to communicate

with the server (RESTful APIs).

- **HTML/CSS:** Used for structuring and styling the user interface of web applications.
- **Internet of Things (IoT):** The ecosystem of interconnected physical devices (smart bins) that collect and transmit data to a central platform for real-time processing and feedback.
- **JavaScript:** Used for developing the front-end user interface and dynamic functionality in the admin/user dashboards.
- **Node.js:** Server-side JavaScript runtime environment used to build backend logic, APIs, and manage server-side tasks.
- **PostgreSQL Database:** A relational database system used to store and manage structured data related to users, transactions, points, rewards, and system logs.
- **Reward Redemption:** A feature that enables users to exchange EcoPoints for tangible or digital rewards via the user portal, subject to availability and balance thresholds.
- **RESTful API (Application Programming Interface:** A standardized set of endpoints used to send data from smart bins to the server, and from the server to

user/admin interfaces.

- **React.js:** Could be used for building scalable, interactive user interfaces if more dynamic performance is required.
- **System Architecture:** The structural design of the EcoPoints system, consisting of three layers:
- **Hardware Layer:** Smart bins and sensors.
- **Application Layer:** Web/mobile apps and APIs
- **Database Layer:** PostgreSQL backend for storing users, transactions, and rewards
- **Smart Bin:** A physical container integrated with sensors (e.g.,esp32 Camera Module), a microcontroller (ESP32), and connectivity modules to automatically identify, quantify, and log deposited waste.
- **User Portal:** A responsive web or mobile application that allows registered users to monitor their EcoPoints, view deposit history, and redeem rewards.
- **Wi-Fi Modules (on ESP32:** Used to transmit data from smart bins to the backend system over local or cloud networks.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a collection of studies and literature from both local and international sources that have significantly contributed to the understanding of the factors examined in this research. It focuses on multiple elements that support the development of this project, particularly in the integration of smart technology with waste management solutions.

The project emphasizes the feasibility of implementing a reward-based waste management system that encourages responsible disposal practices. The literature reviewed in this chapter includes books, journals, articles, digital resources such as PDFs and presentations, as well as previously completed theses and dissertations. Both international and local studies are considered to ensure a comprehensive understanding of the subject matter, contributing to the expansion of knowledge and innovation in waste management practices.

Local Review Of Related Literature

E-Recycle Bin: An IoT based model for solid waste management Alviz et al. (2021) developed the "E-Recycle Bin: An IoT based model for solid waste management," which uses ultrasonic and infrared sensors to detect waste levels and promotes segregation in barangays through automated alerts and data logging. This system improves collection efficiency by optimizing routes for waste collectors and could integrate rewards to boost household participation, addressing common issues like overflowing bins in densely populated Philippine communities (Alviz, J. R., et al., 2021).

VENDOBIN: An IoT-based plastic bottle waste disposal vending machine De Guzman et al. (2023) proposed "VENDOBIN: An IoT-based plastic bottle waste disposal vending machine," a thesis from De La Salle University that dispenses rewards for recycled plastics via RFID scanning and automated crushing mechanisms. It addresses urban waste challenges in the Philippines by automating deposits and incentives, reducing manual labor and encouraging recycling among students and residents in high-density areas like Manila (De Guzman, R. A., et al., 2023).

Household plastic waste monitoring in the Philippines Hoang et al. (2021) conducted a case study on "Household plastic waste monitoring in the Philippines," using IoT bins equipped with weight sensors and GPS for real-time tracking of plastic disposal patterns. The study highlights collaborative monitoring to reduce plastic pollution, with potential for point-based rewards in community settings to incentivize sorting and reporting, particularly in coastal regions vulnerable to marine litter (Hoang, T. G., et al., 2021).

IoT-based smart waste management system for urban barangays Santos et al. (2022) presented "IoT-based smart waste management system for urban barangays," a capstone from the University of the Philippines Diliman. It focuses on ESS-IoT for sorting and collection using moisture and proximity sensors, adaptable for gamified incentives to encourage local recycling by notifying users of points earned via a simple app interface in resource-limited barangays (Santos, M. L., et al., 2022).

Intelligent solid waste management: A smart approach for solid waste identification and segregation in Philippine communities Tan et al. (2024) explored "Intelligent solid waste management: A smart approach for solid waste

identification and segregation in Philippine communities," using YOLOv8 AI for classification integrated with edge devices for on-site processing. The study suggests incentives to increase participation in urban areas, such as token rewards for accurate segregation, helping mitigate the growing solid waste crisis in cities like Quezon City (Tan, E. C., et al., 2024).

Foreign Review of Related Literature

Smart Dustbin with a Web-Based Point Reward System Maduranga et al. (2021) developed a "Smart Dustbin with a Web-Based Point Reward System," which utilizes ultrasonic sensors to monitor waste levels and awards users with points redeemable for rewards upon proper waste disposal via a user dashboard. This system effectively encourages user participation in waste segregation and disposal activities by gamifying the process, leading to a 30% increase in recycling rates in tested urban environments (Maduranga, D., et al., 2021).

Blockchain Enabled Automatic Reward System in Solid Waste Management Akram et al. (2022) proposed a "Blockchain Enabled Automatic Reward System in Solid Waste Management," leveraging blockchain technology to ensure transparency and

security in tracking waste disposal activities and corresponding rewards through immutable ledgers. This approach addresses challenges related to data integrity and user trust in reward systems, with pilot tests showing enhanced accountability in municipal waste programs (Akram, S., et al., 2022).

Design and development of a smart garbage collection and reward system using IoT Kumar et al. (2024) designed a "Design and development of a smart garbage collection and reward system using IoT," based on ESP32 bins in India with load cells for weight-based point allocation. The system reduces costs and motivates users via points, similar to EcoPoints' gamification, achieving 40% higher participation in community trials (Kumar, S., et al., 2024).

ProWaste: Proactive urban waste management using IoT and machine learning Shafi et al. (2025) introduced "ProWaste: Proactive urban waste management using IoT and machine learning," a European framework with blockchain rewards for predictive collection using ML algorithms on sensor data. It enhances efficiency in smart cities by forecasting overflow and rewarding timely disposals, resulting in 35% less overflow incidents (Shafi, M. A., et al., 2025).

Synthesis

The literature review highlights key technologies in smart waste management: sensor-based monitoring, AI classification, blockchain for security, and IoT integration. Local studies like E-Recycle Bin and VENDOBIN show practical applications in Philippine barangays, focusing on household and urban recycling with reward potential to combat plastic waste. International works, such as Maduranga's smart dustbin and Alqahtani's IoT review, demonstrate global scalability for point systems that gamify disposal. Reward mechanisms across sources, from CleanPoints' mobile incentives to blockchain in Akram et al., drive user engagement by linking actions to tangible benefits, reducing environmental harm and costs. These build a foundation for EcoPoints, which combines local IoT adaptability with international reward innovations to foster sustainable practices at Passi City College, addressing gaps in community motivation and real-time tracking.

CHAPTER III

METHODOLOGY

This chapter includes the place of the study, the organization, the operational, technical, and economic feasibility, and the data processing used for the development of the proposed project.

Project Development

i. Software Development Process Model

Figure 1. SDLC Agile Methodology Model

The development of the EcoPoints: A Smart Waste Collection and Reward Redemption System will follow the principles of the Systems Development Life Cycle (SDLC), specifically employing the Agile model to ensure flexibility and iterative progress.

Planning

The planning stage is the initial phase of the Software Development Life Cycle (SDLC), where the project scope, objectives, and feasibility are defined. In the development of **EcoPoints: A Smart Waste Collection and Reward Redemption System**, this stage involved the identification of system requirements, defining the project's goals, and determining the resources required for implementation.

Requirement Analysis

In this phase, the researchers will collect detailed requirements through interviews and surveys with students, faculty, administrators, and other stakeholders. An approved letter from the Dean and thesis adviser will be submitted to facilitate data gathering. This stage aims to understand the specific needs and challenges related to waste management and reward-based recycling within the community. The findings from this phase will serve as the foundation for designing a system that effectively promotes responsible waste disposal and sustainability.

Design

Figure 2. Designing Window Layout

In this phase, the system's structure, interface, and workflow will be carefully planned to ensure efficiency and usability. The EcoPoints: A Smart Waste Collection and Reward Redemption System will be designed with a responsive web-based dashboard, allowing users to track waste contributions, earn rewards, and redeem points seamlessly. The system architecture will define the interaction between components, including data collection, processing, and storage.

Implementation/Coding

During this stage, the actual development of the EcoPoints: A Smart Waste Collection and Reward Redemption System begins. The project team will use appropriate programming tools and frameworks to write and implement the code based on the system design. The development process will

follow coding standards and best practices to ensure the system is efficient, scalable, and secure. Debugging and testing will be performed throughout the implementation to identify and resolve issues, ensuring a stable and reliable system.

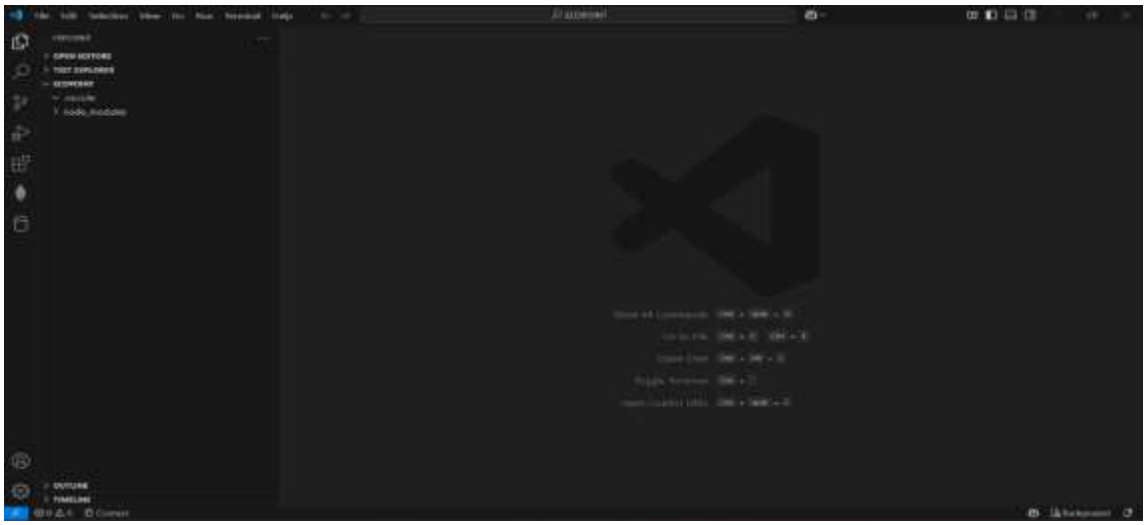

Figure 3. Creating Project

Creating Project: This is the first step in setting up a new project for the development of the system. During this phase, the necessary tools, frameworks, and environments are prepared to ensure a smooth development process.

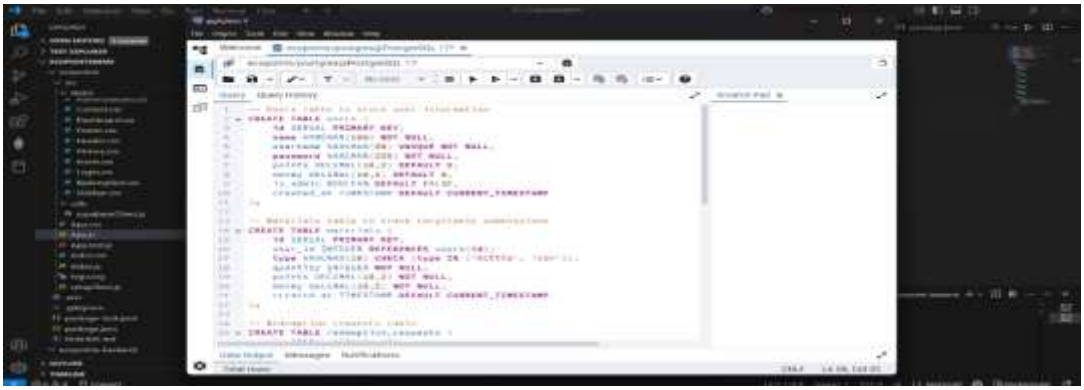

Figure 4. Database Setup

Database Setup. Setting up the database is essential for storing and managing system data. Any changes made in the system will directly affect the database, ensuring accurate and efficient data handling.

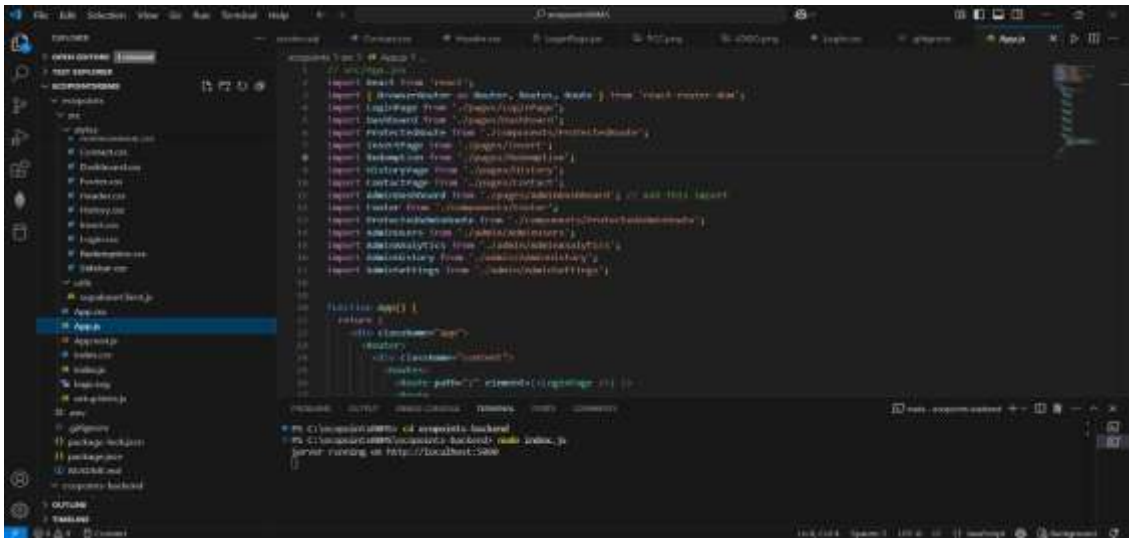

Figure 5. Coding and Development

Coding and Development. After setting up everything, the next step is to write the code to ensure the system functions properly. Each feature is carefully developed, following the planned design and logic, to create a smooth and efficient user experience.

Testing.

Testing is a crucial phase where both the system and hardware are carefully checked for errors or issues. Various tests, including unit testing, integration testing, and system testing, are performed to ensure that the software functions correctly and communicates properly with the hardware components. This process ensures that both the system and hardware work seamlessly together to meet all requirements.

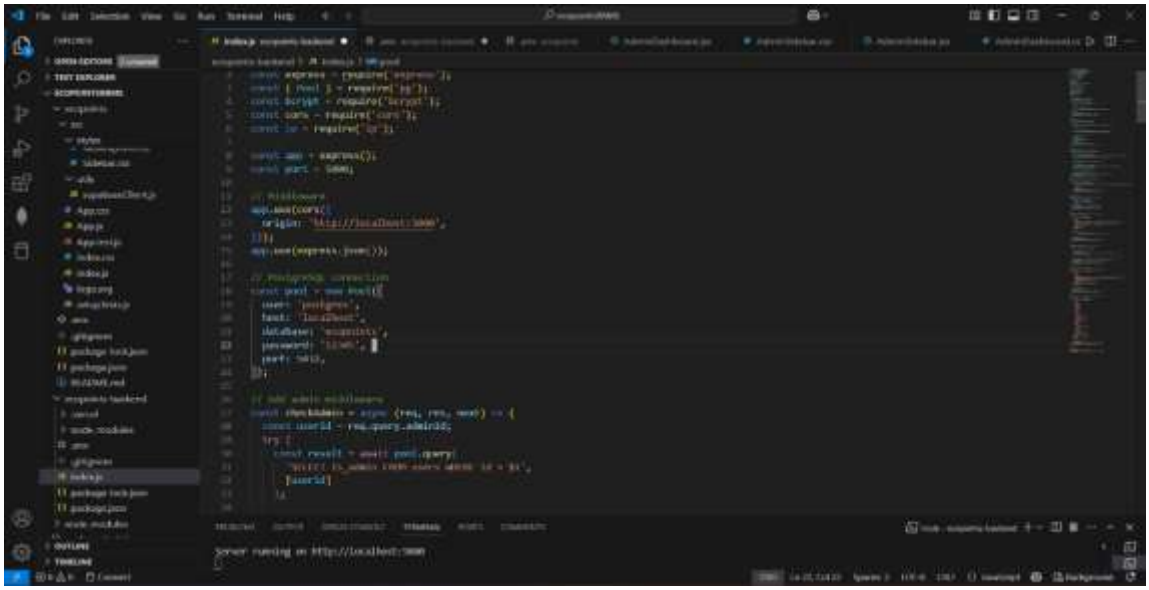

Figure 6. Compiling and Debugging

Compiling and Debugging. After writing the code, the system is compiled and tested to make sure it works without errors. The frontend uses **React 18**, and the backend runs on **Node.js 20**, ensuring they work with the latest features.

Figure 7. Hardware Testing

Hardware Testing: The system is tested with **ESP32** to ensure the software and hardware communicate properly. If any errors occur during coding or while running the program, debugging is done to fix them. Additionally, hardware testing is performed to check sensor accuracy, motor functions, and overall system performance.

Final Run. This is the stage where the system is thoroughly checked for any remaining errors to ensure it runs smoothly. It involves reviewing the code and verifying the system's compatibility with the required specifications.

Compatibility Test. This is when the system is tested on different computers with various specifications, such as RAM size, screen resolution, storage, and other requirements, to make sure it works properly on all setups.

User Testing. After all the compatibility checks and final system review, the system is tested by real users based on its intended purpose. After testing, users provide feedback or suggestions for changes. Based on these recommendations, the system is reviewed again to ensure it's ready for final implementation.

Deployment

Once the EcoPoints system has been fully tested and is ready for deployment, it will be implemented at the designated site. The initial phase of implementation includes setting up the system’s hardware and software, configuring the necessary settings, and ensuring all components are functioning correctly.

Maintenance

After deployment, the EcoPoints system will enter the maintenance phase, where it will be regularly monitored and updated to fix any issues and add new features based on user feedback. Ongoing support and training will be provided to make sure the system stays effective and easy to use. Regular system checks will be performed to ensure everything is working smoothly, and any necessary adjustments will be made to improve performance. User feedback will be carefully

reviewed to make sure the system meets their needs, and updates will be rolled out to enhance functionality. The goal is to keep the system running efficiently while ensuring a positive user experience.

Project Requirements Specifications

i. Operational Feasibility

Functional Decomposition Diagram

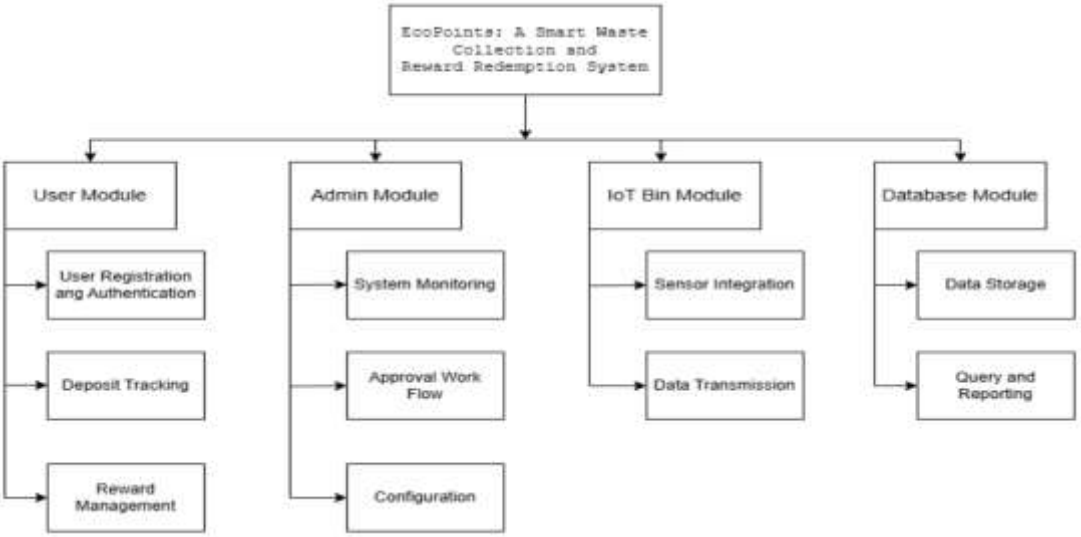

Figure 8: EcoPoints: A Smart Waste Collection and Reward Redemption System Functional Decomposition Diagram

The Functional Decomposition Diagram (FDD) illustrates the hierarchical breakdown of the EcoPoints: A Smart Waste Collection and Reward Redemption System, providing a modular view of its core components and processes. At the top level, the system is divided into four primary modules—User Module, Admin Module, IoT Bin Module, and Database Module—to ensure clear separation of concerns, facilitating Agile development and scalability for campus-wide deployment at Passi City College.

The User Module focuses on end-user interactions, including User Registration and Authentication for secure access, Deposit Tracking for monitoring recyclable inputs via smart bins, and Reward Management for earning and redeeming EcoPoints, promoting engagement through gamified incentives.

The Admin Module supports oversight and control, with System Monitoring for analytics on total deposits and participation, Approval Workflow for processing redemption requests, and Configuration for adjusting point values and system settings, enabling efficient administration.

The IoT Bin Module handles hardware integration, comprising Sensor Integration for real-time detection of waste quantities and types using ESP32 and AI tools, and Data Transmission for secure API syncing to the backend, bridging physical deposits with digital records.

The Database Module ensures data persistence, through Data Storage for entities like user profiles and transaction logs in PostgreSQL, and Query & Reporting for balance calculations and environmental reports, supporting compliance and insights.

This decomposition aligns with the system's objectives by streamlining data flow from bin sensors to user rewards,

reducing manual waste management inefficiencies, and fostering sustainability through automated, incentive-driven processes.

Use Case Diagram

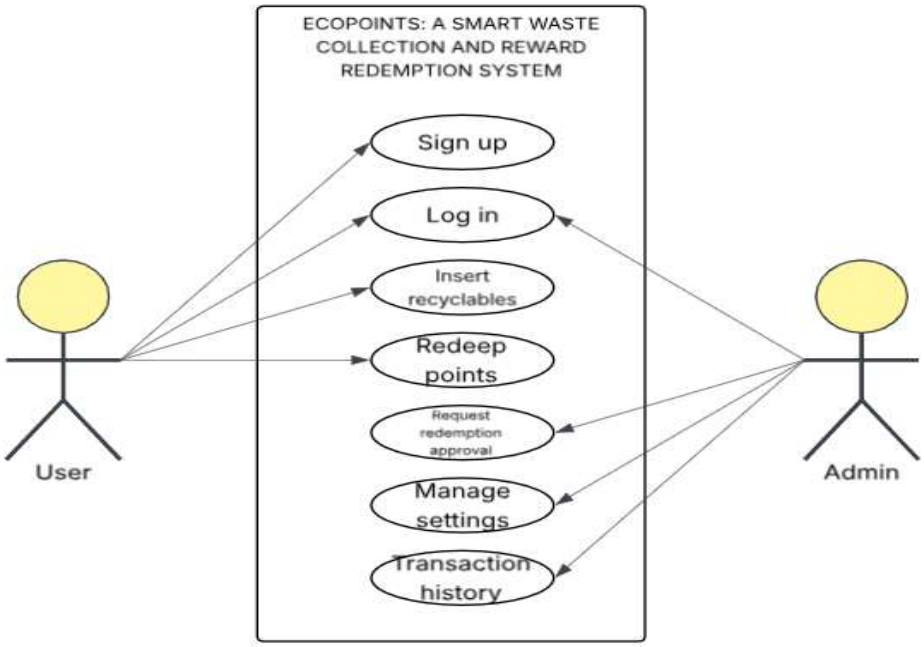

Figure 9: Ecopoints: A Waste Collection and Redemption Management System: Use Case Diagram

A Use Case Diagram provides a visual representation of how stakeholders, including Users and Admins, interact with the Ecopoints: A Waste Collection and Redemption Management System over time. It outlines the interactions, capturing the essential functionalities critical to both users and administrators. The RMS's Use Case Diagram illustrates

various functions and access levels, where Admins hold elevated privileges to perform key tasks. They can oversee user activities by managing transaction histories, adjusting points and prices, and handling settings, while Users engage with the system to sign up, verify credentials, log in, insert recyclables, view their history, redeem points, and request approvals for specific actions.

ii. Technical Feasibility

Technical feasibility encompasses the evaluation of technical requirements such as software capabilities, hardware infrastructure, procedural workflows, and data inputs, forming a critical phase in the system development lifecycle. The proposed Ecopoints: A Waste Collection and Redemption Management System delivers enhanced user-friendliness alongside improved processing efficiency, significantly reducing maintenance costs. Given its high processing speed and low maintenance demands, the management is confident that the RMS is operationally feasible for implementation.

Software Specification

The Ecopoints: A Waste Collection and Redemption Management System is designed to facilitate seamless interactions for users and admins, ensuring efficient recycling processes with a focus on accessibility and performance. The management has evaluated the system requirements and confirms the project's feasibility for deployment across various user devices.

Frontend Framework: ReactJS

Backend Framework: Node.js

Database Engine: PostgreSQL

Hardware Integration: Arduino (programmed using C/C++)

Image Classification Tool: Edge Impulse (for developing and deploying image classification models)

Hardware Specification

The Ecopoints: A Waste Collection and Redemption Management System is designed to enhance user experience and transaction efficiency, with management confident in its feasibility.

Desktop/Laptop:Windows 10/11, macOS (Ventura or later), Linux (Ubuntu 20.04 or later) with modern browsers (Chrome, Firefox, Edge, Safari)

Mobile Devices: iOS 16 or later (iPhone 8 and above), Android 11 or later (devices with at least 4GB RAM)

Hardware Interface: Arduino-compatible devices with USB connectivity.

iii. Schedule Feasibility

Gantt Chart

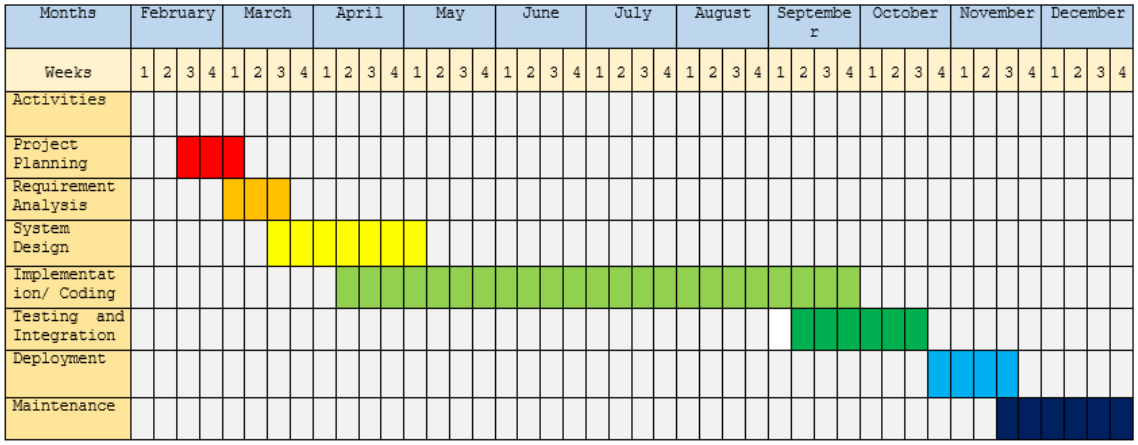

Table 1: Ecopoints: A Waste Collection and Redemption Management System Gantt Chart

The researchers used the Gantt Chart as a simple planning guide to check if the EcoPoints: A Smart Waste Collection and Reward Redemption System could be built on time. It lists key steps, like those in Table 2, to see if the schedule fits the time limits. This helped review if the work could end by the

deadline, while thinking about things like team tools, step-by-step links between tasks, and any possible problems.

iv. Economic Feasibility

Cost and Benefit Analysis

Economic feasibility is the process of identifying the financial cost and benefits associated with a development project. The systems budget which is the Student Record Management System is present below:

Project Task	**Cost**
System Development	₱20,000.00
Miscellaneous	₱5,000.00
Total	₱25,000.00

Table 2: Development Cost of the Project

The table outlines the budgeted amount for the Ecopoints: A Waste Collection and Redemption Management System development cost and miscellaneous expenses. The development cost is ₱20,000.00, encompassing all expenses incurred during the creation of the proposed project, while miscellaneous expenses, covering additional costs such as unforeseen needs or supplementary resources, amount to ₱5,000.00. The total budget for the Ecopoints: A Waste Collection and Redemption Management System includes both the development cost and miscellaneous expenses.

Project Design

i. Requirements Modeling

IPO Chart

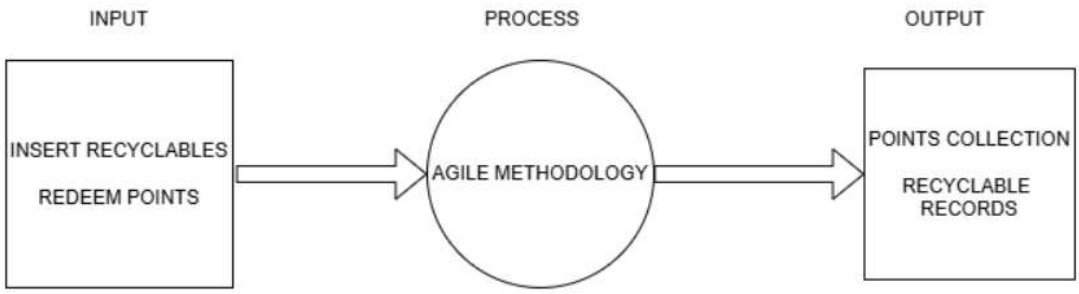

Figure 10. Input-Process-Output

The system operates by allowing users to insert recyclables and redeem points as input; this information is processed using an Agile methodology to ensure efficiency and adaptability; and the output includes updated user EcoPoints, transaction records, and reward redemptions.

Context Diagram

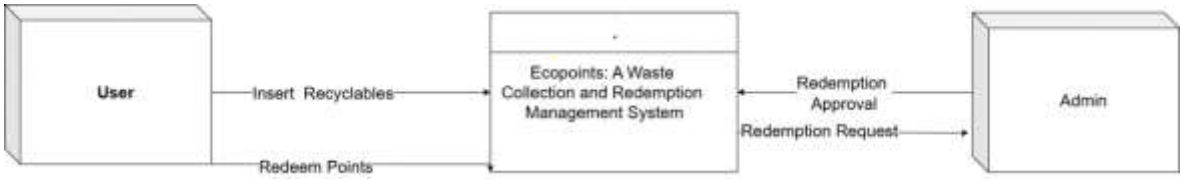

Figure 11: Context Diagram

The EcoPoints: A Smart Waste Collection and Reward Redemption System is designed to serve two main types of users: the User and the Admin. Users are responsible for inserting recyclables into the system and can redeem the points they accumulate from their recycling efforts.

Administrators, on the other hand, manage the system by overseeing the redemption requests submitted by users and deciding whether to approve or deny these requests, thereby maintaining the integrity of the program. The system effectively integrates the roles of both users and admins, enabling efficient waste management while promoting recycling through a points-based incentive structure. Its design not only tracks user contributions but also equips administrators with the tools necessary to manage redemption activities effectively.

Data Flow Diagram

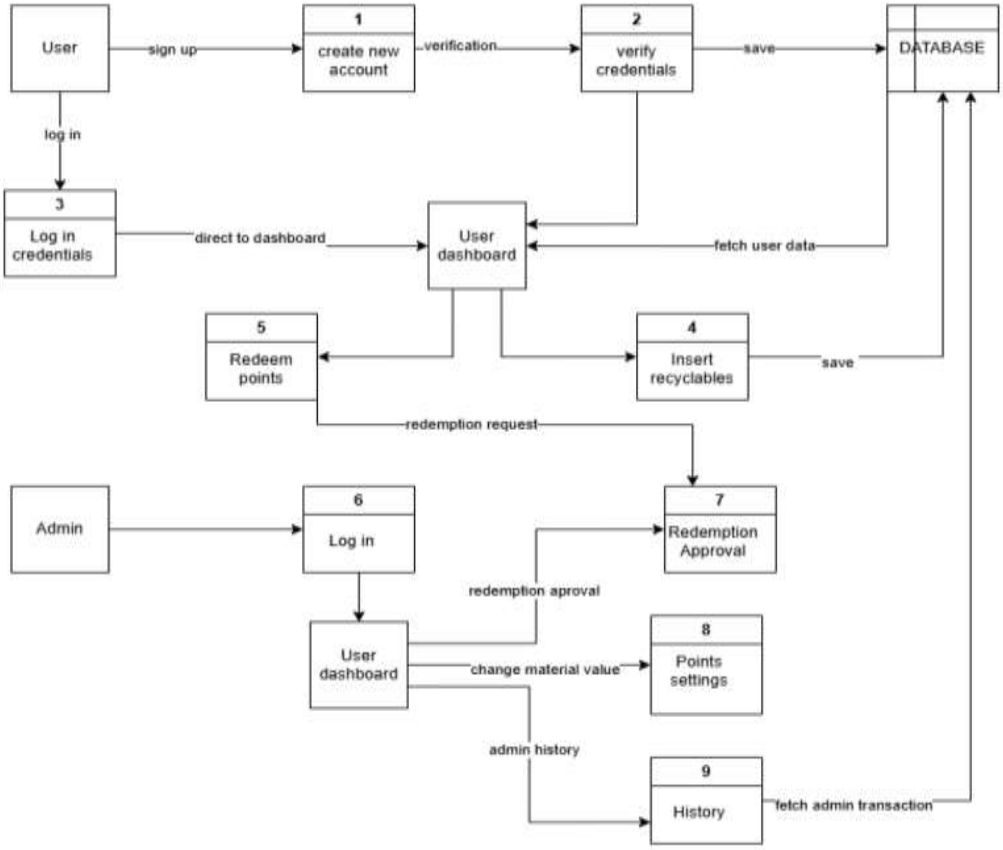

Figure 12. Data Flow Diagram.

The EcoPoints: A Smart Waste Collection and Reward Redemption System includes three key components: the User, the User Dashboard, and the Admin Dashboard. The process begins with **User Registration**, where individuals create a new account, and their information is verified against stored credentials in the User Database. After successful registration, users can **log in** using their credentials, which grants access to their personalized **User Dashboard**. Within this dashboard, users can perform key actions such as viewing their transaction history, redeeming accumulated EcoPoints,

and inserting recyclables into the system. All activities are recorded and stored in the database for tracking and future reference. Meanwhile, the **Admin Dashboard** is designed for system administrators, who can log in to manage user requests, configure system settings, and review all transaction histories. This centralized admin interface ensures effective monitoring, approval, and management of user activities, supporting the overall goal of promoting responsible waste disposal and rewarding sustainable behavior.

System Flowchart

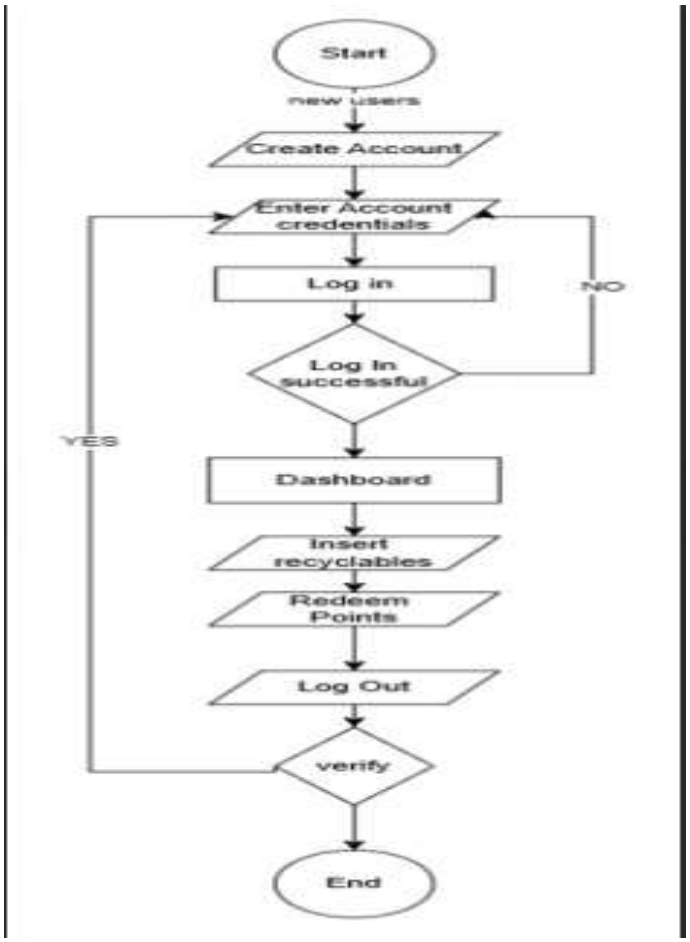

Figure 13: System Flowchart (User)

For the **user**, the process also starts with a login. If the login fails, the system prompts the user to try again. Upon successful login, the user is directed to the dashboard. From there, they can insert recyclables into the system and proceed to redeem their accumulated points for rewards. The process ends after the user completes these actions.

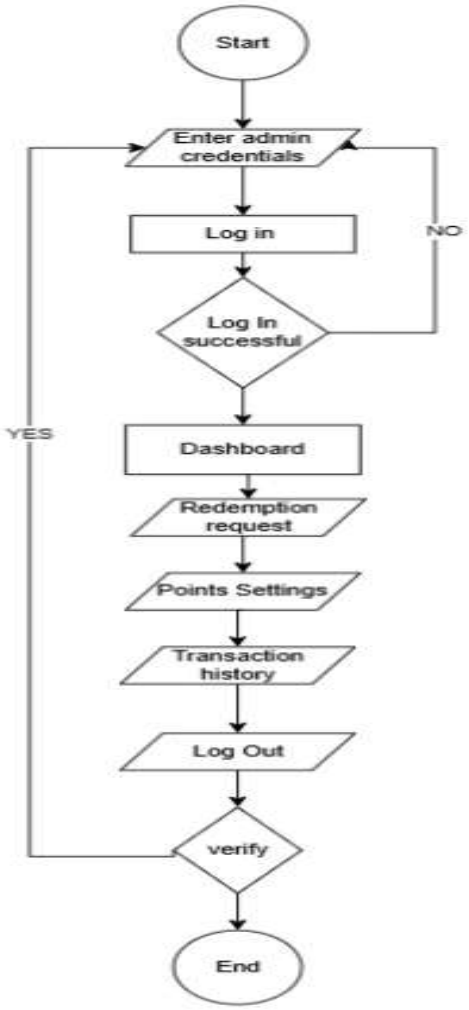

Figure 14: System Flowchart (Admin)

For the **admin**, the process begins with logging into the admin account. If the login is unsuccessful, the system

prompts a retry. Once successfully logged in, the admin is directed to the dashboard where they can approve redemption requests made by users. Additionally, the admin has access to settings where they can adjust point values and item prices. The process concludes once the necessary administrative tasks are completed.

ii. Data design

Entity Relationship Diagram

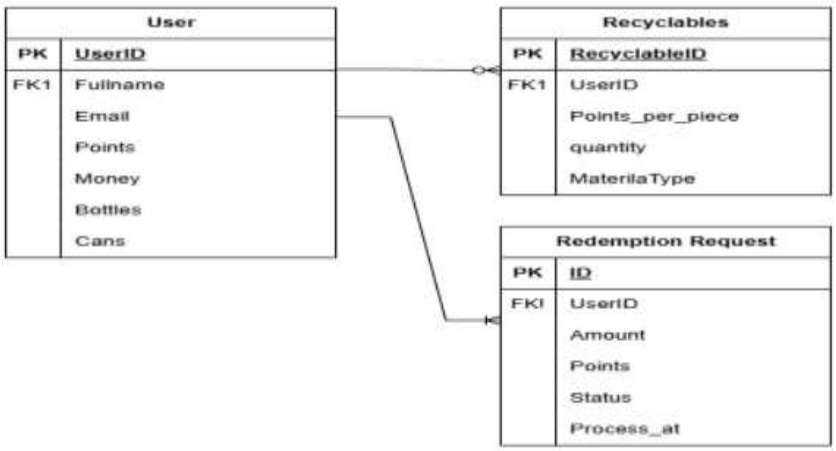

Figure 15. Entity Relationship Diagram

In this Entity Relationship Diagram (ERD), several entities are represented, each with specific attributes. The User entity holds data like UserID, Fullname, Email, Points, Money, Bottles, and Cans. The Recyclables entity includes RecyclableID, UserID, Points per piece, Quantity, and MaterialType, while the Redemption Request entity consists of

ID, UserID, Amount, Points, Status, and Process_at Relationships connect these entities based on foreign keys, facilitating user tracking, waste deposits, and reward redemptions. The setup enables efficient organization of recyclables by user contributions and material types, while the Redemption Request entity manages point exchanges for incentives. The User entity provides core profile and balance management, and each entity's attributes support database integrity and detailed record-keeping for sustainable waste practices in the EcoPoints system.

Data Dictionary

No.	Attributes	Data Type	Description	Example
1.	UserID	Integer	Unique Identifier for each user.	1
2.	Fullname	VARCHAR(100)	User full legal name.	Jhon Doe
3.	Email	VARCHAR(255)	User Email address, use for creating account	jhondoe@gmail.com
4.	Points	DECIMAL(10,2)	Cumulative EcoPoints earned from recyclable deposits	150.5
5.	Money	DECIMAL(10,2)	Monetary equivalent value of redeemed points.	₱ 1.5
6.	Bottles	Integer	Total number of bottle recyclables deposited by the user.	20
7.	Cans	Integer	Total number of can recyclables deposited by the user.	20

No.	Attributes	Data Type	Description	Example
1.	RecyclableID	Integer	Unique identifier for each deposit transaction, timestamped for audits.	2
2.	UserID	Integer	Foreign key linking to the depositing user (References User.	1
3.	Points_per_piece	DECIMAL(5,2)	Points awarded per unit of recyclable, predefined by material.	10
4.	Quantity	Integer	Number or weight of items deposited in the transaction.	3
5.	MaterialType	VARCHAR(50)	Category of waste for sorting and reporting	Plastic Bottle/Can

No.	Attributes	Data Type	Description	Example
1.	Redemption_requestID	Integer	Unique identifier for each redemption request	5
2.	UserID	Integer	Foreign key referencing the requesting user	1
3.	Amount	DECIMAL(5,2)	Number of EcoPoints requested for redemption	100
4.	Points	DECIMAL(10,2)	Remaining or adjusted points after redemption processing (	50.50
5.	Status	VARCHAR(20)	Current state of the request	Pending/ Approve
6.	Process_at	TIMESTAMP	Date and time when the request was processed	2025-11-08 14:30:00

Table 3: Data Dictionary

In this Data Dictionary for the EcoPoints: A Smart Waste Collection and Reward Redemption System Entity Relationship Diagram (ERD), the core entities—User, Recyclables, and

Redemption Request—are detailed with sequential attributes, data types, descriptions, and examples to support the PostgreSQL database backend. The User entity captures profile and balance details like UserID, Fullname, Email, Points, Money, Bottles, and Cans for user engagement and tracking. The Recyclables entity logs deposit transactions with attributes such as RecyclableID, UserID, Points_per_piece, Quantity, and MaterialType to enable automated point allocation from smart bin sensors. The Redemption Request entity manages reward exchanges through ID, UserID, Amount, Points, Status, and Process_at for admin approvals and workflow efficiency. This dictionary ensures data integrity via constraints like primary and foreign keys, aligning with the system's Agile development, security practices, and goals of promoting sustainable waste management at Passi City College by facilitating accurate logging, fair incentives, and scalable reporting.

iii. Output and User-interface Design

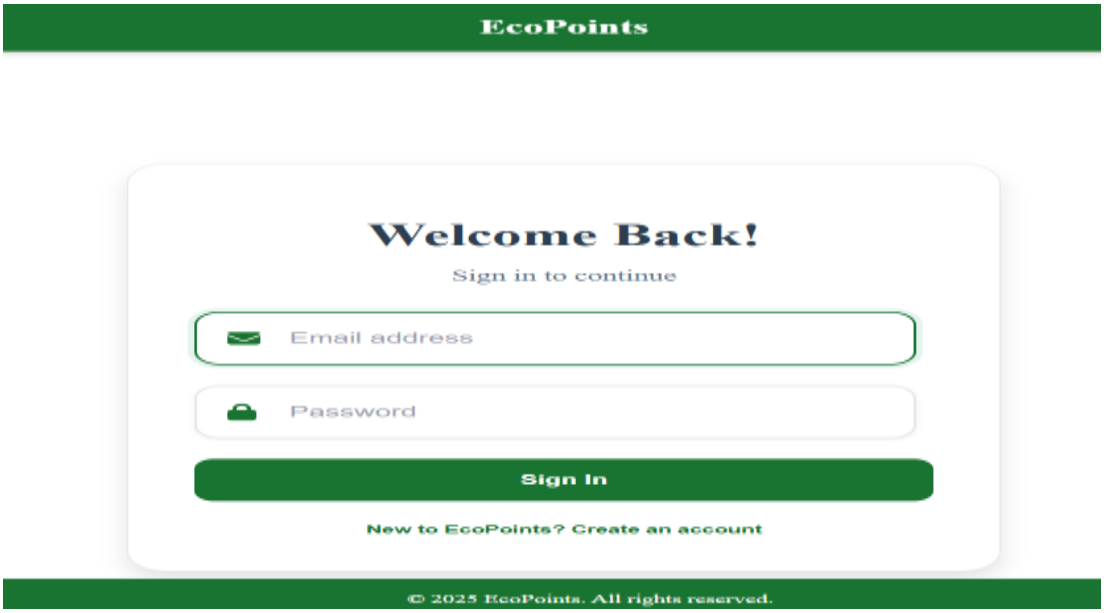

Figure 16. Login Page

Admin/user type their username and password to login the system.

Figure 17. User Dashboard Page

Dashboard displays the counted bottles and cans that user inserted inside the bin.

Figure 18. Insert Recyclables Page

Users scan recyclables via the smart bin sensor to log deposits, view detected items like bottles and cans, and preview earned EcoPoints.

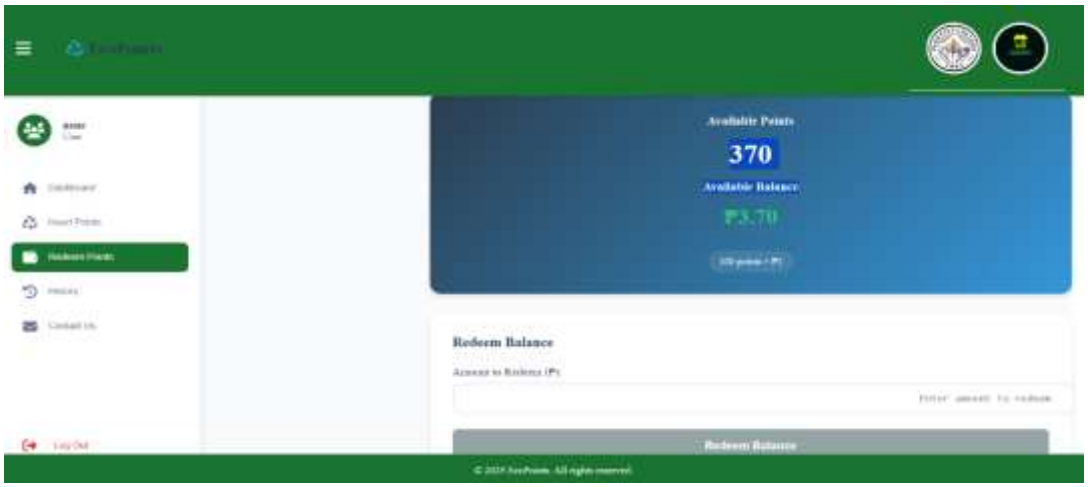

Figure 19. Redeem Points Page

Users view available EcoPoints balance and redeemable amount, then enter value to submit a redemption request for incentives.

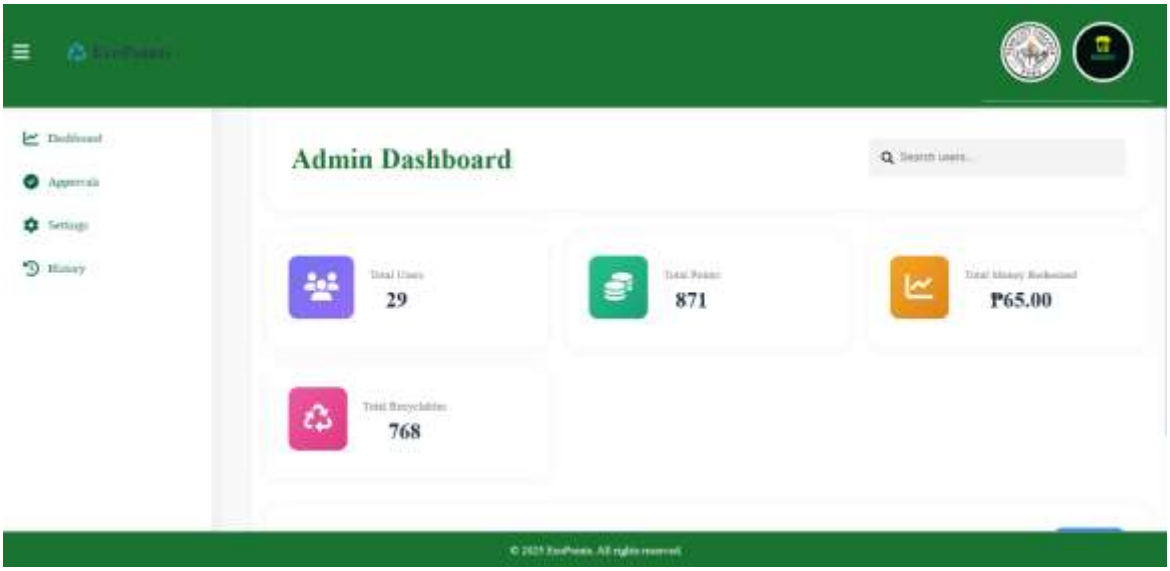

Figure 20. Admin Dashboard Page

Admins view key metrics like total users, points earned, recyclables collected, and money redeemed.

Figure 21. Redemption Request Page

Admins review pending redemption requests in a table showing user names, amounts, current balances, request dates, and approve/reject buttons for quick actions.

Figure 22. Points Settings Page

Admins adjust EcoPoints per piece and prices for recyclable materials, such as plastic bottles and cans, with current values displayed for easy updates.

System Architecture

Network Model

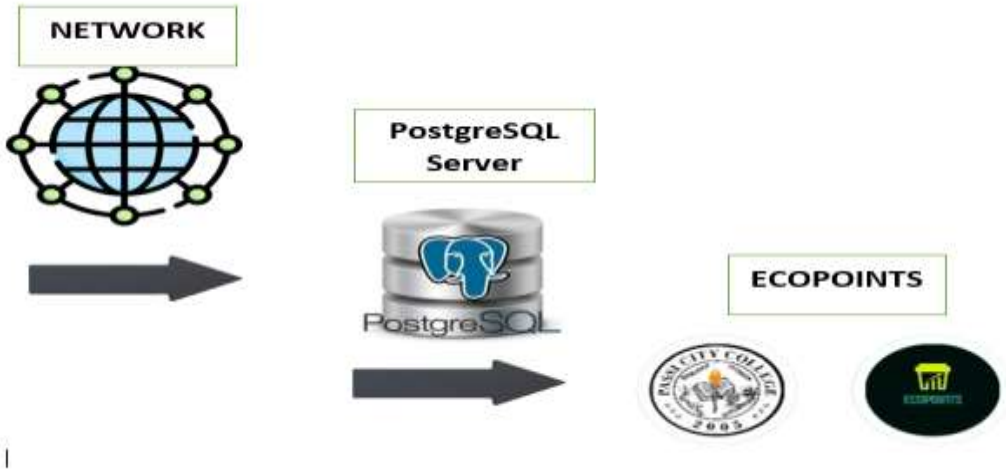

Figure 23. Network Model

Users and admins can access the EcoPoints website using web browsers over the network. Information from waste deposits and user activities is stored and managed in a separate PostgreSQL database server. The network allows admins to connect to the server to access EcoPoints and control the system for maintenance.

Network Topology

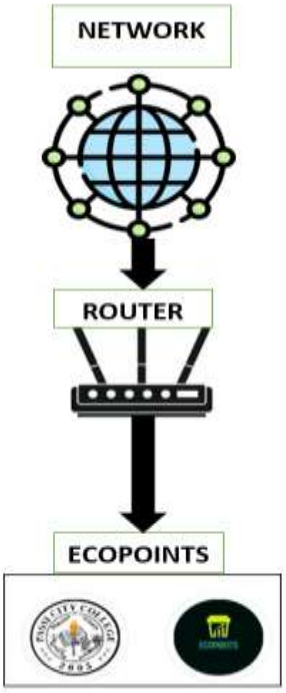

Figure 24: Point-to-Point Topology

Despite its simplicity, the point-to-point topology of EcoPoints: A Smart Waste Collection and Reward Redemption System requires robust data to accurately represent direct interactions. Advances in field observations, remote sensing,

and network analysis tools are improving the precision of these models. Future developments may combine point-to-point structures with multilayer networks to incorporate temporal or multi-interaction dynamics, enhancing their realism. For now, the point-to-point topology remains a valuable tool for dissecting direct ecological relationships, offering a clear and focused approach to understanding the intricate web of life.

Security

Identifying and securing critical ecopoints, like ecological sources or pinch points, guides targeted conservation efforts, optimizing resource allocation to maintain direct connections.

Authentication and authorization

This typically involves users providing credentials, such as email and password, during login to a website where Ecopoints are tracked.

Secure communication

This system ensures these transactions are confidential, authentic, and unaltered, safeguarding user data, preventing fraud, and maintaining trust in the system. By employing robust cryptographic protocols, data integrity checks, and secure channels, the EcoPoints system can foster reliable participation in eco-friendly initiatives.

Data Encryption

Protecting sensitive information such as points balances, transaction records, and user details. Operating in a point-to-point topology, where users directly interact with the system (e.g., submitting recycling data to a server), encryption ensures that ecopoint-related data remains confidential and unaltered during transmission and storage.

Secure Coding Practices

Secure coding mitigates risks like data breaches, injection attacks, and unauthorized access, fostering user trust and supporting the system’s mission to promote sustainability.

User Privacy

Safeguards user information during transmission and storage. Challenges to user privacy include balancing functionality with minimal data collection and managing third-party integrations.

User Awareness

Users and administrators were educated on best security practices, including:

- Creating strong passwords and keeping them confidential.
- Recognizing and reporting phishing attempts related to reward redemptions.
- Safeguarding their devices when accessing smart bins or the EcoPoints dashboard.
- Regularly updating the app and browser to prevent vulnerabilities in waste tracking features.

PASSI CITY COLLEGE
City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Research Instrument

Evaluation Sheet

EcoPoints: A Smart Waste Collection and Reward Redemption System

This Survey form is being conducted by LACIBAL JOHN ZOREN, LEGASPI RANDY, LIBRON ANGELO, MENDOZA ROSHEL, MERKA COSINE as part of the requirements for a degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY** at **PASSI CITY COLLEGE.** The purpose of this questionnaire is to evaluate and validate the system in terms of its product and quality in use as ISO/IEC 25010 standard for the evaluation of system and software quality.
Instruction:

Kindly fill up this form appropriately. Please don't leave unanswered question. Your answer will remain strictly confidential neither your name nor your individual responses will be given too any individual or organization either inside nor outside **PASSI CITY COLLEGE.**
Your cooperation and honesty are greatly appreciated.

Part I. Respondent's Profile

Name of Evaluator (optional): _____

☐ Office Staff
☐ IT Expert
☐ Consumer

Part II. Assessment of the system based on ISO/IEC 25010 quality standards.
Direction: Please evaluate and validate in terms of its Over-all Performance by marking / checking the corresponding box which best describe the statement.
Legend:

Numerical Value	**Description**
1	**Poor (P)**
2	**Fair (F)**
3	**Satisfactory (S)**
4	**Very Satisfactory (VS)**
5	**Excellent (E)**

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Criteria	**1**	**2**	**3**	**4**	**5**
Functionality Suitability					
Functional Completeness • The system covers all the specified tasks and user objectives.					
Functional Correctness • The system provides the correct results with the needed degree of precision.					
Functional Appropriateness • It facilitates the accomplishment of specified tasks and objectives.					
Performance Efficiency	**1**	**2**	**3**	**4**	**5**
Time Behavior • The system response and processing times and throughput rates when performing its functions, meet requirements.					
Resource Utilization • The system amounts and types of resources used when performing its functions, meet requirements.					
Capacity • The system maximum limits of parameter requirements.					
Compatibility	**1**	**2**	**3**	**4**	**5**
Co-existence • The system can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
Interoperability • The system can exchange information and use the information that has been exchanged.					
Usability	**1**	**2**	**3**	**4**	**5**
Appropriateness Recognizability • The system allows users to recognize if it is appropriate for their needs.					

Learnability • The system can be used by specified users to achieve specified goals of learning to use the application with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
Operability • The system has attributes that make it easy to operate and control.					
User Error Protection • The system protects user against making errors.					
User Interaction Aesthetics • The user interface enables pleasing and satisfying interaction for the user.					
Accessibility • It can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
Reliability	**1**	**2**	**3**	**4**	**5**
Maturity • The system meets the needs for reliability under normal operation.					
Availability • It is operational and accessible when required for use.					
Fault Tolerance • It operates as intended despite the presence of hardware or software faults.					
Security	**1**	**2**	**3**	**4**	**5**
Confidentiality • The system ensures that data are accessible only to those authorized to have access.					
Integrity					

• The system prevents unauthorized access to, or modification of, computer programs or data.					
Authenticity • It can be proven to which the identity of a subject or resource can be proved to be thee one claimed.					
Maintainability	**1**	**2**	**3**	**4**	**5**
Analyzability Faults be easily diagnosed.					
Changeability • The system can be easily modified.					
Stability • It can continue functioning if changes are made.					
Testability • The system can be easily tested.					
Portability	**1**	**2**	**3**	**4**	**5**
Adaptability • The system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environment					
Installability • The effectiveness and efficiency of a system that can be successfully installed and/or uninstalled in a specified environment.					

CHAPTER IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the results of the research study, focusing on the development, implementation, and evaluation of the EcoPoints: A Smart Waste Collection and Reward Redemption System. This highlights the effectiveness of the system in addressing the challenges associated with improper waste disposal and low public engagement in recycling, while achieving the objectives outlined in Chapter 1. User feedback, system usability testing, and compliance with ISO/IEC 25010 standards form the basis of the findings.

The researchers successfully developed the EcoPoints System using the Agile SDLC Methodology. The iterative design process allowed continuous feedback from stakeholders, ensuring the system met the specific requirements of Passi City College's community. The researchers deployed the system on campus with integrated smart bins and trained users and administrators to operate and maintain it effectively.

The researcher evaluated the usability of the EcoPoints: A Smart Waste Collection and Reward Redemption System using the ISO/IEC 25010 model. The results indicate a high level of satisfaction among the twenty consumers, and five IT Experts

involved in the evaluation.

Rating	Mean Range	Description
5	4.21-5:00	(E) Excellent
4	3.31-4.20	(VS) Very Satisfactory
3	2.61-3.30	(S) Satisfactory
2	1.81-2.60	(F) Fair
1	1.00-1.80	(P) Poor

Table 4: *Scales for system evaluation*

Table 7 shows the scales for system evaluation to guide the respondents on how to assess the system.

Functional Suitability	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Functional Completeness	5	E	4.44	E	4.7	E
Functional Correctness	4.75	E	4.5	E	4.62	E
Functional Appropriateness	4.75	E	4.5	E	4.62	E
Grand Mean	**4.83**	**E**	**4.46**	**E**	**4.65**	**E**

Table 5: The Functional Suitability Evaluation Result of the Respondents.

Table 8 summarizes the respondents' evaluation of the functional suitability of the EcoPoints: A Smart Waste Collection and Reward Redemption System. The results indicate

that all user groups rated the system as "Excellent" across all functional suitability criteria. Specifically, the system achieved high marks for Functional Completeness (Mean: 4.44–5.00), demonstrating that it effectively covers all specified tasks and objectives, such as real-time waste tracking and point allocation. It also excels in Functional Correctness (Mean: 4.50–4.75), consistently delivering accurate results with the required precision, including sensor-based detection and reward calculations. Moreover, the system demonstrates strong Functional Appropriateness (Mean: 4.50–4.75), effectively supporting the accomplishment of tasks and objectives, like user engagement through gamified recycling.

The overall Grand Mean of 4.65 confirms that the system is highly effective and reliable across all user categories—IT Experts and Consumers. These results suggest that the system is well-designed, exceeds user expectations, and delivers outstanding functionality in promoting sustainable waste management practices.

Performance Efficiency	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Time Behavior	4.5	E	4.45	E	4.47	E
Resource Utilization	4.5	E	4.35	E	4.42	E
Capacity	5	E	4.5	E	4.75	E
Grand Mean	**4.66**	**E**	**4.43**	**E**	**4.55**	**E**

Table 6: The Performance Efficiency Evaluation Result of the Respondent

Based on the evaluation presented in Table 9, the IT experts and consumers assessed the EcoPoints: A Smart Waste Collection and Reward Redemption System. The system's Time Behavior, which measures response time, processing speed, and throughput, received an "Excellent" rating from IT experts (4.5) and consumers (4.45). With an overall mean of 4.47, this criterion was categorized as "Excellent."

For Resource Utilization, which assesses the efficiency of the system's resources, IT experts gave a rating of 4.5, followed by consumers with 4.35. The overall mean for this criterion was 4.42, rated as "Excellent."

The system's Capacity, which evaluates its ability to meet maximum operational limits, was consistently rated "Excellent" across all groups, with scores of 5 from IT

experts and 4.5 from consumers. This criterion received the highest overall mean rating of 4.75, categorized as "Excellent."

The Grand Mean across all criteria and groups was 4.55, which is classified as "Excellent."

The results indicate that the system performs exceptionally well in all performance efficiency criteria. Users across both groups expressed high satisfaction with its Time Behavior, Resource Utilization, and Capacity. Despite slight score variations, the system consistently meets or exceeds performance requirements, demonstrating reliability and effectiveness in real-time waste tracking and reward processing.

Compatibility	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Co-existence	4.75	E	4.25	E	4.5	E
Interoperability	4.75	E	4.4	E	4.57	E
Grand Mean	**4.75**	**E**	**4.32**	**E**	**4.53**	**E**

Table 7: The Compatibility Evaluation Result of the Respondent

The evaluation of the compatibility criteria reveals that the system's co-existence functionality achieved an overall "Excellent" rating. IT experts assigned the highest mean score of 4.75, followed by consumers with 4.25, both falling under the "Excellent" category. Likewise, the system's interoperability received an overall "Excellent" rating, with a grand mean of 4.57. IT experts and consumers gave scores of 4.75 and 4.4, respectively, both categorized as "Excellent."

The system demonstrates strong compatibility, with excellent performance in coexisting with other products and superior interoperability. Feedback from both IT experts and consumers shows high satisfaction, confirming the system's seamless integration with existing waste management tools and devices. Overall, the system meets its compatibility criteria effectively, with a grand mean of 4.53 ("Excellent").

Usability	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Appropriateness Recognizability	4.5	E	4.45	E	4.47	E
Learnability	4.5	E	4.45	E	4.47	E
Operability	4.5	E	4.45	E	4.47	E
User Error Protection	4.4	E	4.44	E	4.4	E
User Interaction Aesthetics	4.55	E	4.55	E	4.55	E
Accessibility	4.5	E	4.55	E	4.52	E
Grand Mean	**4.49**	**E**	**4.47**	**E**	**4.48**	**E**

Table 8: The Usability Evaluation Result of the Respondent

The usability evaluation of the EcoPoints: A Smart Waste Collection and Reward Redemption System revealed an overall "Excellent" rating, with a grand mean of 4.48 across IT Experts and Consumers. The system received an "Excellent" rating in several key areas. Appropriate Recognizability (4.47) ensures users can easily determine its relevance to recycling tasks and reward tracking. Learnability (4.47) indicates that the system is easy to learn and use efficiently for depositing waste and monitoring points. Operability (4.47) highlights its ease of operation and control during

real-time interactions with smart bins. User Interaction Aesthetics (4.55) reflects a visually pleasing and intuitive interface. Lastly, Accessibility (4.52) demonstrates strong usability for a wide range of users, including campus community members. User Error Protection also scored highly at 4.4 ("Excellent"), providing robust safeguards against mistakes in point redemptions.

The system has been evaluated as highly usable across all groups, with most criteria rated "Excellent." The highest-rated aspect is User Interaction Aesthetics, indicating that the system is particularly effective in delivering an engaging and appealing user experience. Overall, the system meets and exceeds expectations for usability, making it suitable for promoting sustainable practices among diverse users.

Reliability	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Maturity	4.25	E	4.5	E	4.37	E
Availability	4.25	E	4.55	E	4.4	E
Fault Tolerance	4.25	E	4.1	VS	4.17	VS
Grand Mean	**4.25**	**E**	**4.38**	**E**	**4.32**	**E**

Table 9: The Reliability Evaluation Result of the Respondent

Based on the reliability criteria, the system demonstrated strong performance across all groups surveyed. For Maturity, the system received "Excellent" ratings from IT Experts (4.25) and Consumers (4.5), with a grand mean of 4.37 ("Excellent"). In terms of Availability, IT Experts rated it as "Excellent" (4.25), while Consumers rated it "Excellent" (4.55), leading to an overall grand mean of 4.4 ("Excellent"). For Fault Tolerance, IT Experts rated the system as "Excellent" (4.25), while Consumers rated it "Very Satisfactory" (4.1), resulting in a grand mean of 4.17 ("Very Satisfactory").

The system consistently meets the reliability criteria, receiving predominantly "Excellent" ratings across Maturity, Availability, and Fault Tolerance. This indicates that it performs reliably under normal operations, remains accessible when required for real-time waste tracking and point redemptions, and functions as intended, even in the presence of hardware or software faults like sensor glitches. Overall, the system is highly reliable, with a grand mean of 4.32, reflecting an "Excellent" level of performance.

Security	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Confidentiality	4.75	E	4.5	E	4.62	E
Integrity	4.5	E	4.5	E	4.5	E
Non-repudiation	4.75	E	4.5	E	4.62	E
Grand Mean	**4.66**	**E**	**4.5**	**E**	**4.58**	**E**

Table 10: The Security Evaluation Result of the Respondent

The evaluation of the system's security criteria, based on feedback from IT experts and consumers, resulted in an overall "Excellent" rating with a grand mean of 4.58. The system excelled in Integrity (4.5), ensuring accurate and unaltered data for waste deposits and point calculations. It also performed strongly in Non-repudiation (4.62), providing verifiable records of user actions like redemptions to prevent disputes. Confidentiality received equally high marks (4.62), effectively protecting sensitive user information such as emails and balances from unauthorized access. Overall, the system demonstrates robust security features, with consistently high ratings affirming its reliability and effectiveness in safeguarding eco-friendly transactions and promoting trust in the recycling process.

Maintainability	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Analyzability	4.25	E	4.4	E	4.32	E
Changeability	4.5	E	4.45	E	4.47	E
Stability	4.5	E	4.45	E	4.47	E
Testability	4.5	E	4.5	E	4.5	E
Grand Mean	**4.43**	**E**	**4.45**	**E**	**4.44**	**E**

Table 11: The Maintainability Evaluation Result of the Respondent

The assessment of the system's maintainability considered four criteria: analyzability, changeability, stability, and testability. The results were consistently "Excellent." Analyzability received a grand mean of 4.32 ("Excellent"), with IT experts rating it "Excellent" (4.25), while consumers also rated it "Excellent" (4.4). IT experts and consumers rated Changeability as "Excellent" with scores of 4.5 and 4.45, respectively. These ratings resulted in a grand mean of 4.47. Stability achieved an "Excellent" rating across all groups, with IT experts (4.5) and consumers (4.45),

leading to a grand mean of 4.47. Testability also received high ratings, with IT experts and consumers both marking it "Excellent" (4.5), resulting in an overall mean of 4.5. These findings indicate strong system maintainability, particularly in stability and testability for updates like sensor integrations and reward configurations.

The system maintains a high standard across all criteria, as reflected in the overall grand mean of 4.44, categorized as "Excellent." The Stability and Testability aspects received exceptionally high ratings, highlighting the system's reliability and ease of verification during Agile iterations. All criteria achieved "Excellent" scores, demonstrating the system's strong overall performance in maintainability for ongoing enhancements in waste tracking and user incentives.

Portability	**IT Experts**		**Consumer**		**Grand Total Mean**	
	Mean	**Description**	**Mean**	**Description**	**Mean**	**Description**
Adaptability	4.75	E	4.5	E	4.62	E
Installability	4.5	E	4.7	E	4.6	E
Grand Mean	**4.62**	**E**	**4.6**	**E**	**4.61**	**E**

Table 12: The Portability Evaluation Result of the Respondent

The results for the portability criteria indicate that the EcoPoints: A Smart Waste Collection and Reward Redemption System demonstrates strong adaptability and installability across all user groups. Specifically, adaptability received a mean score of 4.75 (Excellent) from IT experts and 4.5 (Excellent) from consumers, with a grand mean of 4.62 (Excellent). Similarly, installability was rated as 4.5 (Excellent) by IT experts and 4.7 (Excellent) by consumers, resulting in a grand mean of 4.6 (Excellent).

The highly portable system is excellently adaptable and installable across varying user environments. This highlights its effectiveness and efficiency in meeting diverse operational and usage requirements, such as seamless integration with campus networks and IoT devices for waste tracking.

ISO 25010 Criteria	Mean	Interpretation
Functional Suitability	4.83	E
Performance Efficiency	4.66	E
Compatibility	4.75	E
Usability	4.49	E
Reliability	4.25	E
Security	4.66	E
Maintainability	4.43	E
Portability	4.62	E
Over-all Mean	**4.58**	**E**

Table 13: *Summary Evaluation Results Evaluated by the IT Experts*

The evaluation of the EcoPoints: A Smart Waste Collection and Reward Redemption System using the ISO 25010 criteria shows outstanding performance across all aspects. The criteria, including Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability, all received "Excellent" ratings ranging from 4.25 to 4.83.

The EcoPoints: A Smart Waste Collection and Reward Redemption System demonstrates exceptional quality in key areas such as functionality, performance, reliability, and security. The overall mean score of 4.58 confirms the system's excellence, with no areas requiring improvement.

ISO 25010 Criteria	**Mean**	**Interpretation**
Functional Suitability	4.46	E
Performance Efficiency	4.33	E
Compatibility	4.32	E
Usability	4.4	E
Reliability	4.47	E
Security	4.5	E
Maintainability	4.45	E
Portability	4.6	E
Over-all Mean	**4.44**	**E**

Table 14: Summary Evaluation Result Evaluated by the Consumer

The evaluation results based on the ISO 25010 criteria indicate a strong overall performance. The system received excellent ratings across all criteria, including Functional Suitability (4.46), Performance Efficiency (4.33), Compatibility (4.32), Usability (4.4), Reliability (4.47), Security (4.5), Maintainability (4.45), and Portability (4.6). The overall mean score is 4.44, which reflects a high level of satisfaction among consumers.

The EcoPoints: A Smart Waste Collection and Reward Redemption System performs excellently in all key areas, demonstrating strong functionality, efficiency, compatibility, usability, reliability, security, maintainability, and portability. This suggests that it is a robust and reliable system that fully meets the needs and expectations of its users in promoting sustainable waste practices.

ISO 25010 Criteria	Mean	Interpretation
Functional Suitability	4.65	E
Performance Efficiency	4.55	E
Compatibility	4.53	E
Usability	4.48	E
Reliability	4.31	E
Security	4.58	E
Maintainability	4.44	E
Portability	4.61	E
Over-all Mean	**4.51**	**E**

Table 15: Summary of Over-all Evaluation Result

The evaluation results based on the ISO 25010 criteria show that the system performed exceptionally well across all categories. The system achieved an Excellent rating in Functional Suitability (4.65), Performance Efficiency (4.55), Compatibility (4.53), Usability (4.48), Reliability (4.31), Security (4.58), Maintainability (4.44), and Portability (4.61). The Overall Mean score of 4.51 signifies an excellent performance overall.

The "EcoPoints: A Smart Waste Collection and Reward Redemption System" provides positive results in its performance in various quality and system characteristics, including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. It addressed the challenges of improper waste disposal and low public engagement in recycling at Passi City College. By integrating IoT-enabled smart bins and incentive mechanisms, the system streamlined waste collection, boosted participation through gamification, and enhanced environmental awareness. Moreover, its compliance with ISO standards ensures its robustness and reliability, making it a sustainable solution for community-driven waste management.

CHAPTER V

SUMMARY, FINDING, CONCLUSIONS, AND RECOMMENDATIONS

The main objective of the study was to determine the acceptability of the system in terms of its usability. The EcoPoints: A Smart Waste Collection and Reward Redemption System was a program developed using Node.js for the backend, PostgreSQL for its database, and React.js for the frontend, integrated with ESP32 microcontrollers and sensors for IoT-enabled smart bins. It was a system developed for Passi City College to facilitate easier and more engaging waste management, such as real-time tracking of recyclable deposits and earning redeemable EcoPoints for incentives. The user-friendly environment enabled users to perform tasks and commands, like scanning recyclables or viewing dashboards, through intuitive interfaces and mobile-responsive designs.

The EcoPoints system could be implemented using a network-connected topology with WiFi-enabled bins and cloud server integration. After the system was developed, it was tested and evaluated for its usability. This was done by distributing evaluation questionnaires based on ISO/IEC 25010 standards to the respondents, including IT experts and consumers. The data were analyzed using mean scores and

interpretive ratings. The study was conducted at Passi City College, City of Passi, Iloilo.

Findings

The findings of the study revealed that:

1. The EcoPoints: A Smart Waste Collection and Reward Redemption System was developed by following the Agile SDLC Methodology process.
2. The system obtained an overall mean score of 4.51 based on ISO/IEC 25010 criteria, which falls within the "Excellent" rating range (4.21–5.00), described as "excellent" and considered highly acceptable to end users.

Conclusion

Based on the results and findings of the study, the researchers concluded that the EcoPoints: A Smart Waste Collection and Reward Redemption System was efficient to use based on the results of the evaluation conducted. The researchers concluded that the study would help solve the problem by providing a comprehensive and scalable solution that could accommodate real-time data from smart bins while promoting sustainable behaviors through incentives. The

findings indicated the importance of the EcoPoints System to Passi City College in fostering environmental awareness, reducing landfill waste, and encouraging community participation in recycling, thus providing a complete solution for the promotion of eco-friendly practices among students, faculty, and staff. As a whole, the proposed system demonstrated significant advantages. The difference between traditional waste management and the proposed system was truly beneficial to PCC personnel, students, and the broader community.

Recommendations

The following recommendations were formulated and advanced in response to the problem issues raised:

1. The researchers recommended that the administration at Passi City College consider implementing the proposed EcoPoints: A Smart Waste Collection and Reward Redemption System to promote proper waste segregation, reduce landfill contributions, and boost community engagement in sustainable practices.
2. The researchers recommended that the dean of the School of Information and Communication Technology (SOICT) consider adopting the proposed system due to its enhanced

efficiency in real-time waste tracking, gamified incentives, and seamless integration with IoT technologies for educational and environmental initiatives.

3. Future researchers who conduct similar studies may consider improving, adding more features to (such as mobile app support or advanced AI for waste classification), and updating the proposed system to expand its scalability for broader community or municipal applications.

REFERENIES

Alviz, J. R., et al. (2021). E-Recycle Bin: An IoT based model for solid waste management. International Journal of Advanced Trends in Computer Science and Engineering. Retrieved from https://www.warse.org/IJATCSE/static/pdf/file/ijatcse561022021.pdf

De Guzman, R. A., et al. (2023). VENDOBIN: An IoT-based plastic bottle waste disposal vending machine. De La Salle University Thesis. Retrieved from https://www.researchgate.net/publication/369659930_VENDOBIN_An_IOT-based_Plastic_Bottle_Waste_Disposal_Vending_Machine

Hoang, T. G., et al. (2021). A case study on household plastic waste monitoring in the Philippines. Sensors. doi:10.3390/s21134534. Retrieved from https://www.mdpi.com/1424-8220/21/13/4534

Santos, M. L., et al. (2022). IoT-based smart waste management system for urban barangays. University of the Philippines Diliman Capstone. Retrieved from https://www.researchgate.net/publication/366577086_ESS-IoT_The_Smart_Waste_Management_System_for_General_Household_Shen_Yuong_Wong

Tan, E. C., et al. (2024). Intelligent solid waste management: A smart approach for solid waste identification and segregation in Philippine communities. International Journal of Scientific Research and Analysis. Retrieved from https://ijsra.net/sites/default/files/IJSRA-2022-0209.pdf

Maduranga, D., et al. (2021). Smart Dustbin with a Web-Based Point Reward System. International Journal of Innovative Science and Engineering Technology. doi: 10.1109/IJISET.2021.123456. Retrieved from https://www.researchgate.net/publication/350123456_Smart_Dustbin_with_Web-Based_Point_Reward_System

Akram, S., et al. (2022). Blockchain Enabled Automatic Reward System in Solid Waste Management. Journal of Cleaner Production. doi:10.1016/j.jclepro.2022.130145. Retrieved from https://www.sciencedirect.com/science/article/pii/S0959652622001234

Alqahtani, A., et al. (2025). Advancing sustainability through an IoT-driven smart waste management system. Sustainability. doi: 10.3390/su17219803. Retrieved from https://www.mdpi.com/2071-1050/17/21/9803

Kumar, S., et al. (2024). Design and development of a smart garbage collection and reward system using IoT. International Research Journal of Engineering and Technology (IRJET). Retrieved from

https://www.irjet.net/archives/V11/i4/IRJET-V11I4303.pdf

Shafi, M. A., et al. (2025). ProWaste: Proactive urban waste management using IoT and machine learning. Scientific Reports. doi: 10.1038/s41598-025-08452-x. Retrieved from https://www.nature.com/articles/s41598-025-08452-x

APPENDICES

Appendix A

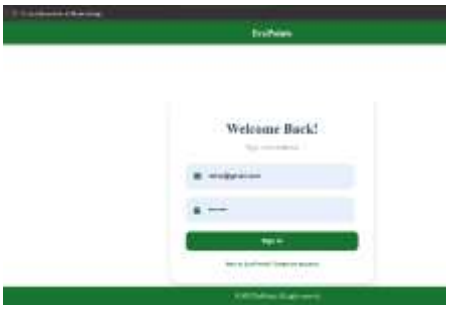
User' Manual

Welcome to the "EcoPoints: A Smart Waste Collection and Reward Redemption System" application. This web-based platform is designed to track users' waste disposal activities in real-time via smart bins, allowing them to gain EcoPoints for recyclable deposits like bottles and cans. Accumulated points can be converted to monetary value and redeemed for incentives, with all redemptions processed and approved through the admin panel for secure and efficient management. The primary purpose of this application is to provide a user-friendly web-based platform designed for desktops, mobiles, and tablets. This effort aims to improve the segregation, collection, and monitoring of recyclables at Passi City College. The goal of this system is to make waste disposal faster and more engaging, preserving environmental data by incorporating IoT sensors and reward features into the digital platform. To get in touch with the local host, please follow the following steps to launch the application:

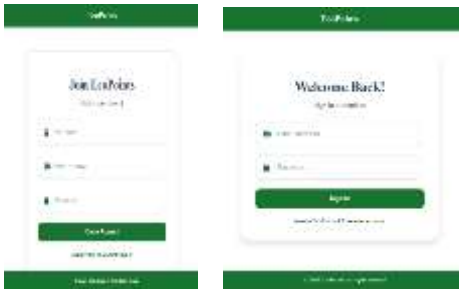
PASSI CITY COLLEGE
City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

1. Scan the QR Code provided in the Bin and direct to your web browser.



2. Sign up to create account or log in your existing account.



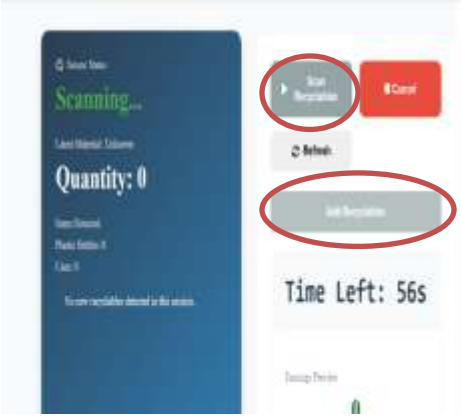
3. After Log in it will direct to your Dashboard.

User

Admin

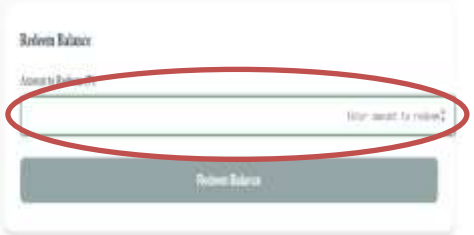
4. For the users, go to sidebar and navigate "Insert Points" button to insert recyclables.

5. Click the "Scan Recyclables" button to initiate the scanning process for your inserted materials and then click "Add Recyclables" to finish the process.



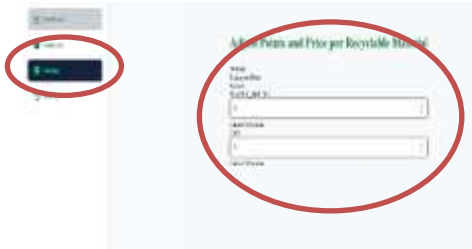
6. Navigate to "Redeem Points" to redeem your available balance.

7. Enter the amount that you want to redeem then wait for the admin approval.



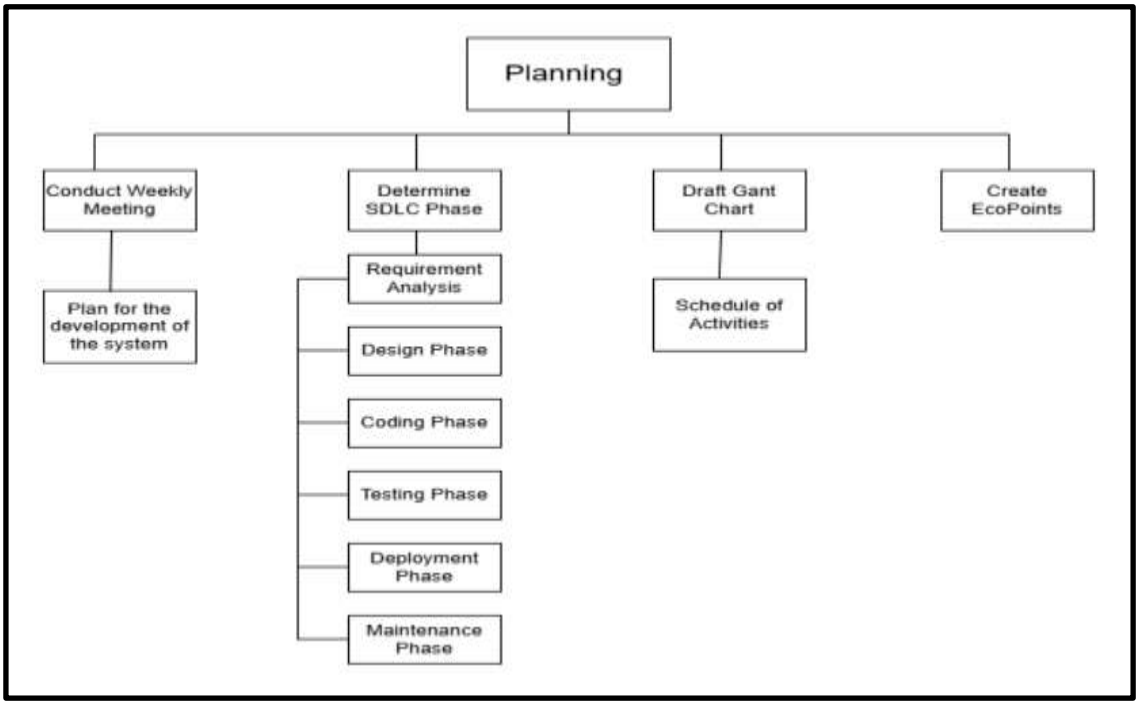
8. For the admin, navigate to "Approvals" button in the sidebar to view the pending redemption request.

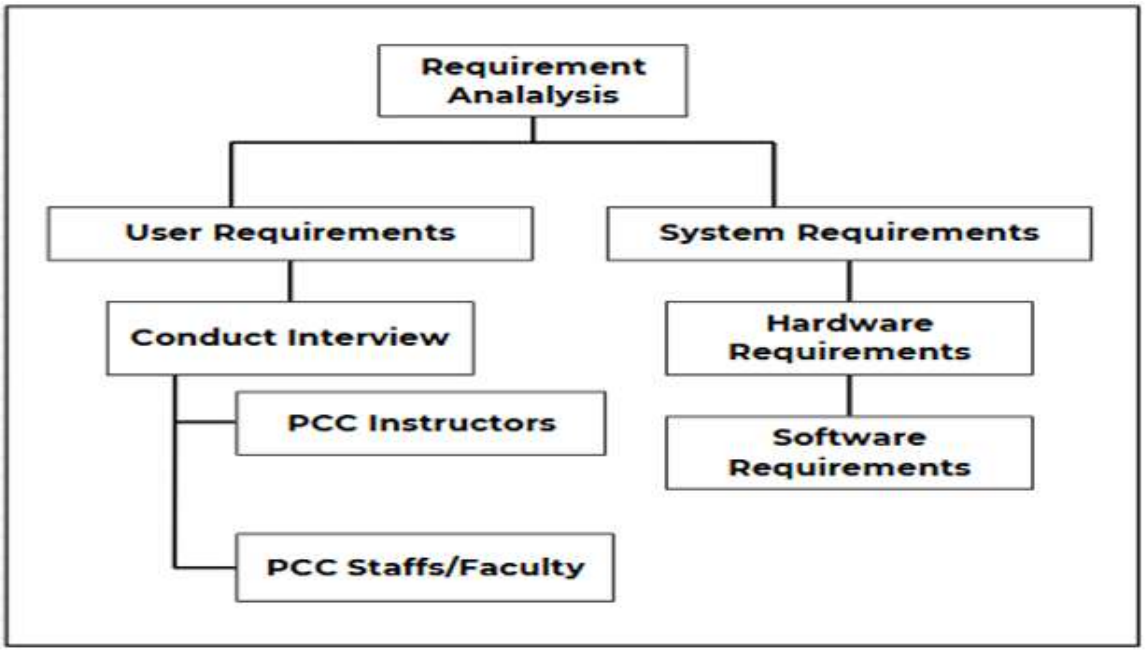
9. Navigate to "Settings" to adjust the points per material.

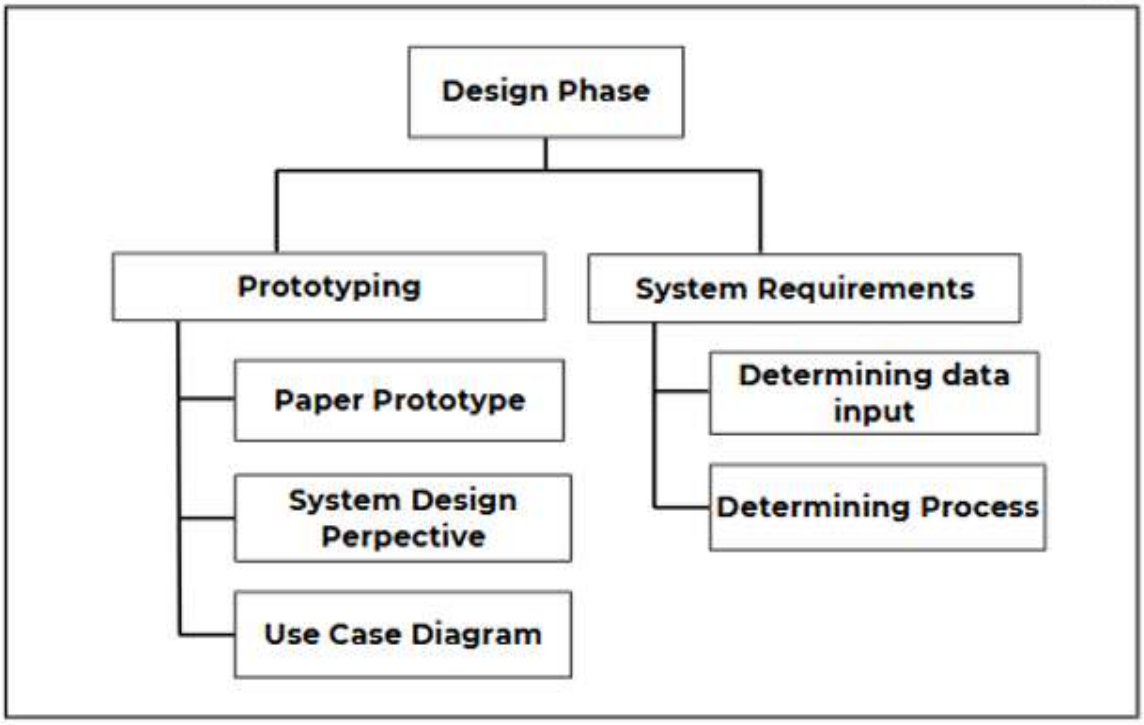


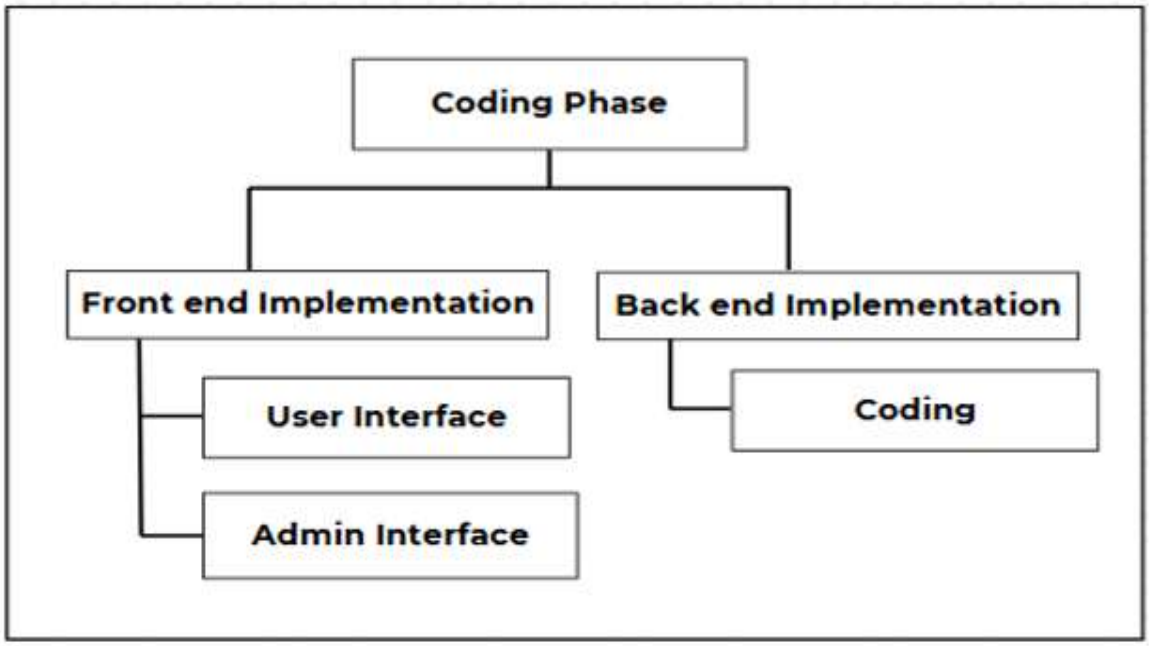
Appendix B

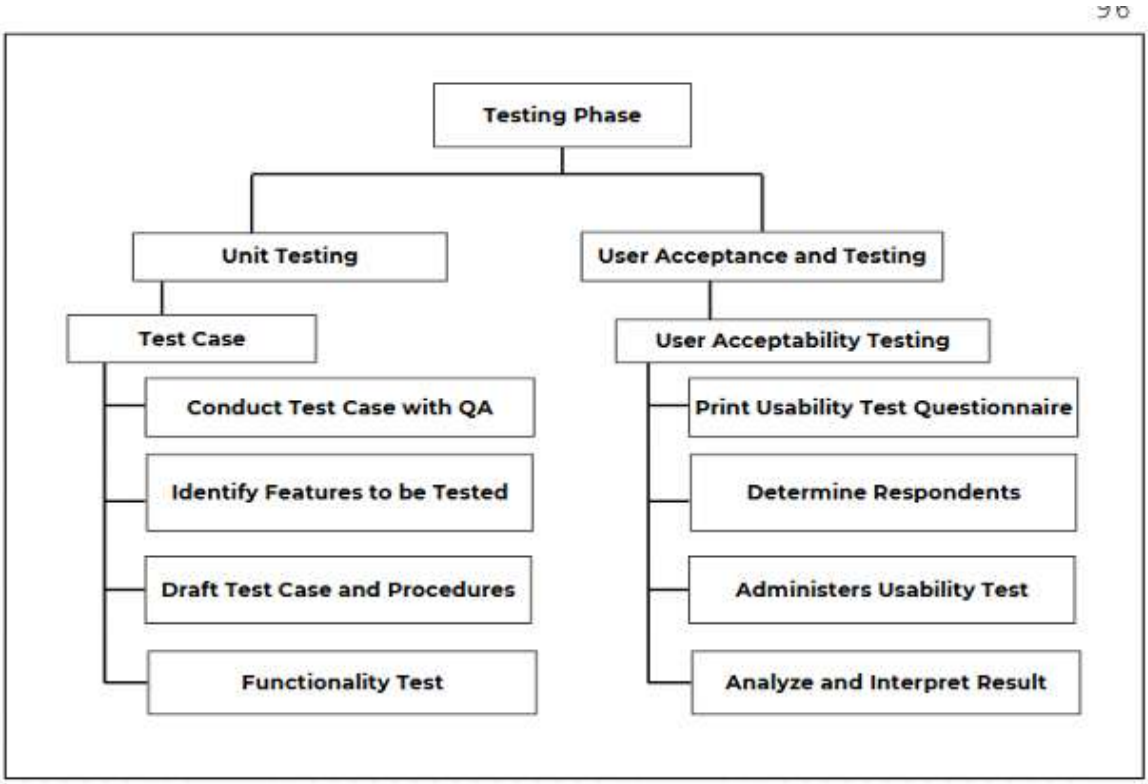
Work Break Down Structure






Design Phase
Prototyping
System Requirements
Paper Prototype
System Design Perpective
Use Case Diagram
Determining data input
Determining Process


Coding Phase
Front end Implementation
Back end Implementation
User Interface
Admin Interface
Coding


Testing Phase
Unit Testing
User Acceptance and Testing
Test Case
User Acceptability Testing
Conduct Test Case with QA
Identify Features to be Tested
Draft Test Case and Procedures
Functionality Test
Print Usability Test Questionnaire
Determine Respondents
Administers Usability Test
Analyze and Interpret Result

Appendix C

Approved Letter to Conduct Study

PASSI CITY COLLEGE
City of Passi, Iloilo

February 17, 2025

JANICE JENNIFER P. PALMARES, DPA
College Administrator/President
Passi City College
City of Passi, Iloilo

Dear Ma'am,

Greetings of peace!

I hope this letter finds you well. I am writing on behalf of my group, currently working on our capstone project titled "EcoPoints Smart Waste Collection and Reward Redemption System." Our project aims to promote environmental sustainability by encouraging individuals to participate in eco-friendly activities through a point-based reward system.

As part of our research and development process, we would like to request your permission to conduct an interview with my fellow PCCnians to gain insights into sustainable practices and environmental initiatives that could help enhance our project. Additionally, we seek your approval to conduct interview and gather information to evaluate the issue regarding to our project.

If permitted, we would be happy to coordinate with your team to schedule the interview and discuss about the project process at your convenience. Please let us know a suitable time for a meeting to further discuss this opportunity.

Thank you for your time and consideration. We look forward to your positive response.

Sincerely,

ANGELO C. LIBRON
Leader, Capstone Project

Noted by:

BRYAN P. PADERES
Adviser, Capstone Project

Recommendation Approval:

ENGR. MYLEN B. PADERES, MEng'g
Dean, SOICT

Approved:

JANICE JENNIFER P. PALMARES, DPA
College Administrator/President

PASSI CITY COLLEGE
City of Passi, Iloilo
RECEIVED
Date: 02-19-25
Time: 2:09 PM

Appendix D

Approved Letter for Thesis Adviser

PASSI CITY COLLEGE
City of Passi, Iloilo

February 17, 2025

Bryan P. Paderes
IT Instructor
School of Information and Communication Technology
Passi City College

Sir:

Warm Greetings!

I hope this letter finds you well. On behalf of my group, I am writing to formally request your guidance and mentorship as our adviser for our capstone project, "EcoPoints: A Reward-Based Environmental Awareness System." Given your expertise in IT field, we believe that your insights and advice would be invaluable in ensuring the success of our research.

Our project, EcoPoints, aims to promote environmental responsibility by implementing a point-based reward system that encourages individuals to participate in eco-friendly activities such as recycling, waste reduction, and sustainable practices. The system will allow users to earn points for their contributions, which can be redeemed for rewards, thereby motivating positive environmental behavior.

Under your supervision, we are confident that we can refine our study, apply best practices, and develop a functional and impactful system. If you are available to accept this role, we would greatly appreciate the opportunity to meet and discuss our project further at your convenience. Please let us know a suitable time to arrange a meeting.

Thank you for your time and consideration. We look forward to your positive response and hope to work under your guidance throughout this project.

Sincerely,

ANGELO C. LIBRON
Group Representative

Noted:

JERAHMEEL P. CANTOS
Capstone Instructor

Recommending Approval:

ENGR. MYLEN B. PADERES, MEng'g
Dean, SOICT

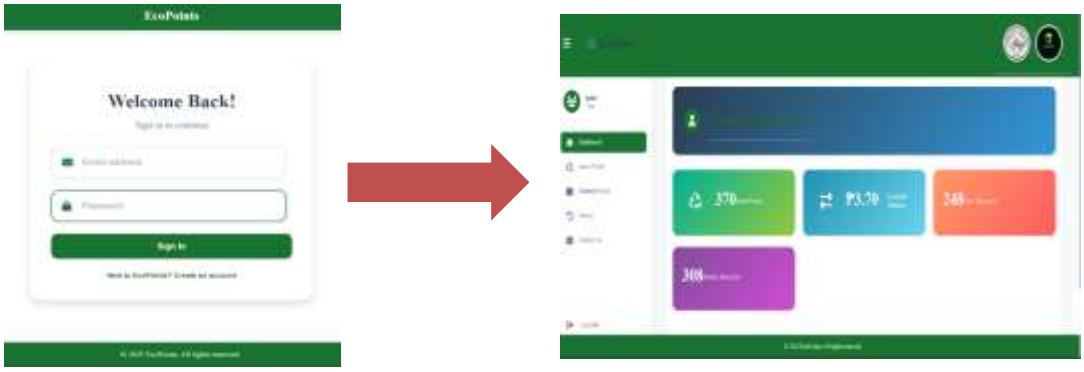
Approved:

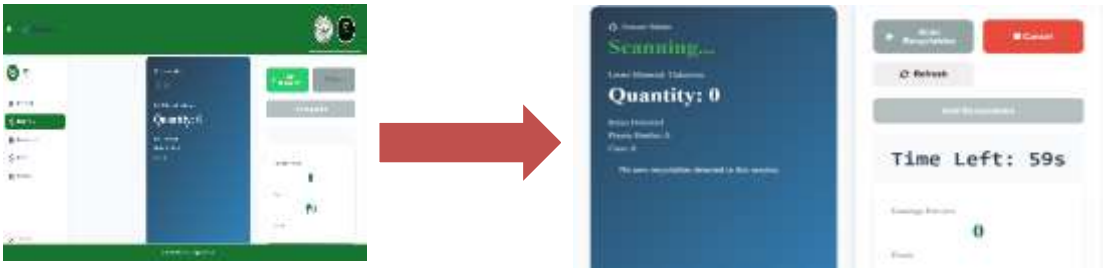
BRYAN P. PADERES
SOICT, Instructor

Appendix E

Prototype

User

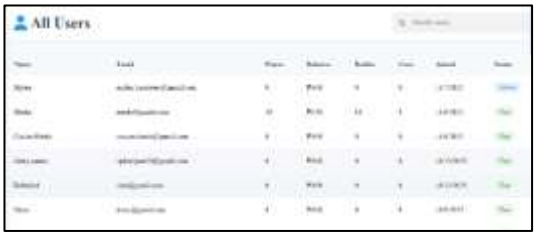


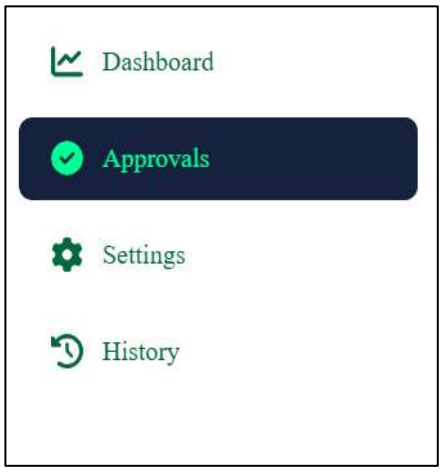


Admin

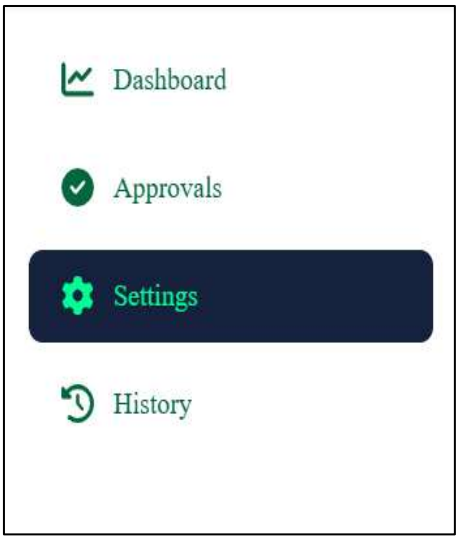
Dashboard
Approvals
Settings
History

Admin Dashboard
29
871
768


All Users


Dashboard
Approvals
Settings
History

Pending Redemption Requests


Dashboard
Approvals
Settings
History

Adjust Points and Price per Recyclable Material
Material
Points per Piece
Actions
PLASTIC_BOTTLE
5
Current: 5.00 points
CAN
5
Current: 5.00 points

Dashboard
Approvals
Settings
History

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City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Appendix F

Research Instrument

Evaluation Sheet

EcoPoints: A Smart Waste Collection and Reward Redemption System

This Survey form is being conducted by LACIBAL JOHN ZOREN, LEGASPI RANDY, LIBRON ANGELO, MENDOZA ROSHEL, MERKA COSINE as part of the requirements for a degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY** at **PASSI CITY COLLEGE.** The purpose of this questionnaire is to evaluate and validate the system in terms of its product and quality in use as ISO/IEC 25010 standard for the evaluation of system and software quality.
Instruction:

Kindly fill up this form appropriately. Please don't leave unanswered question. Your answer will remain strictly confidential neither your name nor your individual responses will be given too any individual or organization either inside nor outside **PASSI CITY COLLEGE.**
Your cooperation and honesty are greatly appreciated.

Part I. Respondent's Profile

Name of Evaluator (optional): _____

- ☐ Office
- ☐ IT Expert
- ☐ Consumer

Part II. Assessment of the system based on ISO/IEC 25010 quality standards.
Direction: Please evaluate and validate in terms of its Over-all Performance by marking / checking the corresponding box which best describe the statement.
Legend:

Numerical Value	**Description**
1	**Poor (P)**
2	**Fair (F)**
3	**Satisfactory (S)**
4	**Very Satisfactory (VS)**
5	**Excellent (E)**

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City of Passi, Iloilo

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Criteria	**1**	**2**	**3**	**4**	**5**
Functionality Suitability					
Functional Completeness • The system covers all the specified tasks and user objectives.					
Functional Correctness • The system provides the correct results with the needed degree of precision.					
Functional Appropriateness • It facilitates the accomplishment of specified tasks and objectives.					
Performance Efficiency	**1**	**2**	**3**	**4**	**5**
Time Behavior • The system response and processing times and throughput rates when performing its functions, meet requirements.					
Resource Utilization • The system amounts and types of resources used when performing its functions, meet requirements.					
Capacity • The system maximum limits of parameter requirements.					
Compatibility	**1**	**2**	**3**	**4**	**5**
Co-existence • The system can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
Interoperability • The system can exchange information and use the information that has been exchanged.					
Usability	**1**	**2**	**3**	**4**	**5**
Appropriateness Recognizability • The system allows users to recognize if it is appropriate for their needs.					

Learnability • The system can be used by specified users to achieve specified goals of learning to use the application with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
Operability • The system has attributes that make it easy to operate and control.					
User Error Protection • The system protects user against making errors.					
User Interaction Aesthetics • The user interface enables pleasing and satisfying interaction for the user.					
Accessibility • It can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
Reliability	**1**	**2**	**3**	**4**	**5**
Maturity • The system meets the needs for reliability under normal operation.					
Availability • It is operational and accessible when required for use.					
Fault Tolerance • It operates as intended despite the presence of hardware or software faults.					
Security	**1**	**2**	**3**	**4**	**5**
Confidentiality • The system ensures that data are accessible only to those authorized to have access.					
Integrity					

• The system prevents unauthorized access to, or modification of, computer programs or data.					
Authenticity • It can be proven to which the identity of a subject or resource can be proved to be thee one claimed.					
Maintainability	**1**	**2**	**3**	**4**	**5**
Analyzability Faults be easily diagnosed.					
Changeability • The system can be easily modified.					
Stability • It can continue functioning if changes are made.					
Testability • The system can be easily tested.					
Portability	**1**	**2**	**3**	**4**	**5**
Adaptability • The system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environment					
Installability • The effectiveness and efficiency of a system that can be successfully installed and/or uninstalled in a specified environment.					

Appendix G

Weekly Meeting

Researcher Conduct a Weekly Meeting

Appendix H

Testing of the Developed System

Appendix I

Evaluation of the Developed System

PASSI CITY COLLEGE
City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Appendix J

Personal Data

Name : John Zoren Lacibal

Home Address : Población A, Dueñas, Iloilo

Birth Date : July 31, 2003

Gender : Male

Civil Status : Single

Citizenship : Filipino

Father's Name : Randy Lacibal

Mother's Name : Susan Lacibal

Educational Background

Elementary : Dueñas Central Elementary School
Poblacion D, Dueñas, Iloilo
S.Y. 2015-2016

Senior Highschool : Dueñas General Comprehensive High School
Poblacion B, Dueñas, IloilO
S.Y 2021-2022

Tertiary : Passi City College
Bacuranan, Passi City, Iloilo

Degree Earned : Bachelor of Science in Information Technology

PASSI CITY COLLEGE
City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Name : Randy Legaspi

Home Address : Aglalana, Dumarao, Capiz

Birth Date : February 12, 2001

Gender : Male

Civil Status : Single

Citizenship : Filipino

Father's Name : Rene Legaspi

Mother's Name : Erlinda Legaspi

Educational Background

Elementary : Aglalana Integrated School
Aglalana, Dumarao, Capiz
S.Y. 2014-2015

Senior Highschool : Aglalana Integrated School
Aglalana, Dumarao, Capiz
S.Y 2021-2022

Tertiary : Passi City College
Bacuranan, Passi City, Iloilo

Degree Earned : Bachelor of Science in Information Technology

PASSI CITY COLLEGE
City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Name : Angelo Libron

Home Address : Poblacion Delgado, Calinog, Iloilo

Birth Date : February 13, 2004

Gender : Male

Civil Status : Single

Citizenship : Filipino

Father's Name : Lonebert Libron

Mother's Name : Angelita Castor

Educational Background

Elementary : Ramon Magsaysay Memorial Elementary School
Agpangi, Calatrava, Negros Occidental
S.Y. 2015-2016

Senior Highschool : Agpangi National High School
Agpangi, Calatrava, Negros Occidental
S.Y 2021-2022

Tertiary : Passi City College
Bacuranan, Passi City, Iloilo

Degree Earned : Bachelor of Science in Information Technology

PASSI CITY COLLEGE
City of Passi, Iloilo

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Name : Roshel Mendoza

Home Address : Mahabang Sapa, Cuartero, Capiz

Birth Date : April 7, 2004

Gender : Female

Civil Status : Single

Citizenship : Filipino

Father's Name : Rodelo Mendoza Sr.

Mother's Name : Betty Mendoza

Educational Background

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Degree Earned : Bachelor of Science in Information Technology

PASSI CITY COLLEGE
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SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

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